

Amplifying mutational profiling of extracellular vesicle mRNA with SCOPE

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Sequencing of messenger RNA (mRNA) found in extracellular vesicles (EVs) in liquid biopsies can provide clinical information such as somatic mutations, resistance profiles and tumor recurrence. Despite this, EV mRNA remains underused due to its low abundance in liquid biopsies, and large sample volumes or specialized techniques for analysis are required. Here we introduce Self-amplified and CRISPR-aided Operation to Profile EVs (SCOPE), a platform for EV mRNA detection. SCOPE leverages CRISPR-mediated recognition of target RNA using Cas13 to initiate replication and signal amplification, achieving a sub-attomolar detection limit while maintaining single-nucleotide resolution. As a proof of concept, we designed probes for key mutations in *KRAS*, *BRAF*, *EGFR* and *IDH1* genes, optimized protocols for single-pot assays and implemented an automated device for multi-sample detection. We validated SCOPE's ability to detect early-stage lung cancer in animal models, monitored tumor mutational burden in patients with colorectal cancer and stratified patients with glioblastoma. SCOPE can expedite readouts, augmenting the clinical use of EVs in precision oncology.

Liquid biopsies present opportunities in precision oncology by providing dynamic molecular information of the entire tumor through minimally invasive and repeatable tests^{1–3}. As such, both researchers and clinicians use liquid biopsies to monitor tumor evolution and heterogeneity⁴. Accessing real-time insights can impact patient care through early lesion detection, minimal residual disease tracking, personalized therapeutic decision-making informed by resistance profiles and tumor recurrence monitoring^{5,6}.

Extracellular vesicles (EVs) are analytical targets within the liquid biopsy arena^{7–9}. These small particles (<1 µm in diameter) carry diverse molecular cargo, including nucleic acids, proteins and metabolites, thus serving as cellular surrogates^{10,11}. Detecting EV messenger RNA (mRNA) can generate rich and actionable clinical information. EV mRNA can reflect somatic driver mutations (for example, *KRAS*^{G12D}

and *BRAF*^{V600E}) that are crucial for tumor initiation and growth¹². Additionally, EVs rarely contain the nuclear proteins that confer drug resistance, but they harbor corresponding mRNA counterparts to inform resistance status^{13–15}. Because EV mRNAs are enclosed within vesicles, they are shielded from nucleases in biofluids, allowing for the isolation of intact, high-quality nucleic acids¹⁶. These collective advantages position EVs as a promising source of nucleic acids, complementing the benefits of circulating tumor DNA (ctDNA).

However, the clinical potential of EVs remains underused, primarily due to technical drawbacks. Most EV RNA is non-coding, and mRNA copy numbers in EV samples can be very low. For instance, even abundant mRNA species, such as *GAPDH*, are only detected at a frequency of one copy per 10⁴–10⁶ EVs, compared to microRNAs at one copy per 10² EVs¹⁷. This disparity has led most proof-of-concept

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assay development to conveniently focus on microRNA detection^{18–20}. Moreover, tumor-derived EVs constitute a minor fraction (<5%) of total circulating EVs²¹. The scarcity of EV mRNAs, combined with the low abundance of tumor-derived EVs, necessitates large sample volumes (over 2 ml of plasma) and sophisticated resources, for example, droplet-digital polymerase chain reaction (PCR) and next-generation sequencing. This diminishes the competitive advantages of EV tests and complicates their integration into routine preclinical and clinical assays.

In the present study, we developed a streamlined, rapid EV mRNA test, drawing inspiration from the CRISPR method. CRISPR technologies are increasingly adopted in molecular diagnostics for their sequence-specific nuclease activity^{22,23}. CRISPR-associated (Cas) proteins become active endonucleases upon recognizing target nucleic acids²⁴. This property has been exploited to amplify signals through the promiscuous cleavage of reporter probes (for example, single-stranded DNAs tagged with a fluorescent dye and quencher pair). However, applying CRISPR assays to EV mRNA proves challenging due to the low abundance of targets. Separate pre-amplification is usually needed to replicate mRNA targets and improve assay kinetics²³. Replication errors and biases inherent in such amplifications can propagate^{25,26}, leading to confounding results. To address this fundamental challenge, we repurposed the Cas activity, where the high-precision Cas system recognizes target mRNA and then triggers its replication in situ, bypassing the need for pre-amplification and associated errors. This robust and efficient strategy could ensure high analytical sensitivity while maintaining sequence specificity, a crucial capability for identifying scarce mRNA targets.

Here we present the Self-amplified and CRISPR-aided Operation to Profile EVs (SCOPE) platform, an integrated assay for accurate EV mRNA detection and monitoring. SCOPE leverages the synergy between the Cas13a machinery and signaling templates. Cas13a first recognizes its target RNA, and then both RNA targets and fluorescent signals are amplified. The high selectivity afforded by Cas13a allows SCOPE to resolve single-nucleotide polymorphisms. The assay further achieves a sensitivity down to a sub-attomolar detection limit, owing to its built-in dual amplification mechanism and further optimizations. We applied SCOPE for various applications, such as detecting early-stage lung cancer in animal models, tracking cancer mutational burden in patients with colorectal cancer (CRC) undergoing standard of care and identifying mutations in glioblastoma multiforme (GBM) for patient stratification. The developed assay can promote preclinical and clinical opportunities to dissect cancer growth processes, identify the emergence of resistance and gauge tumor response to treatments. Ultimately, SCOPE offers to expedite standard clinical and drug trial decisions and enhance the use of EVs in the liquid biopsy space.

Results

Overall SCOPE flow

The SCOPE strategy combines CRISPR recognition with RNA replication within a single assay format. The assay is initiated by introducing samples to an all-in-one SCOPE mixture containing the complex of Cas13a and CRISPR RNA (Cas13a/crRNA), T7 polymerase, signal template and deoxyribonucleotide triphosphate (Fig. 1a). The signal template is crucial for coupling Cas13a/crRNA and T7 polymerase reactions. It is designed explicitly as an RNA–DNA hybrid comprising (1) a fluorescent RNA segment cleavable by Cas13a/crRNA, (2) a fluorescent quencher and (3) a DNA template for target RNA transcription by T7 polymerase (Supplementary Fig. 1).

The major steps of the SCOPE reaction unfold as follows (Fig. 1b and Supplementary Fig. 2). (i) Cas13a/crRNA recognizes and binds to RNA targets. (ii) The complex becomes an active ribonuclease and promiscuously degrades the RNA segment in the signal template. This *trans*-cleavage turns on fluorescent signals by releasing dye molecules

from quenchers and exposes the DNA template complementary to the target RNA. (iii) T7 RNA polymerase attaches to the promoter region of the template and replicates RNA targets. (iv) Unreacted Cas13a/crRNA further recognizes RNA target replicas, reinforcing the overall reaction rather robustly.

The SCOPE mechanism offers important technical advantages for RNA detection. First, it capitalizes on the high selectivity of Cas13a/crRNA complexes^{23,27}. The crRNA design incorporates a strategically placed synthetic mismatch, which empowers Cas13a to discriminate between RNA sequences, even down to single-nucleotide variations²⁷. In SCOPE, this highly specific Cas13a acts as a powerful filter—Cas13a activates target amplification only in the presence of a bona fide target RNA. This strategy enriches the desired target RNA signal within a complex background of other RNAs. Second, SCOPE achieves high sensitivity (low detection limit) through dual amplifications: activated Cas enzymes keep cleaving RNA segments to boost fluorescent signals, and RNA polymerases make multiple copies of the target RNA. Third, the reaction is fast and carried out in a single tube at a constant temperature (40 °C). These features render SCOPE simple and readily applicable across routine laboratory settings (for example, academia, hospitals, primary care clinics and industry).

We streamlined the workflow of detecting RNA targets from EV lysates. Specifically, we simplified RNA extraction and detection steps for clinical EV analysis (Fig. 1c). In this procedure, RNA is swiftly (10 min) extracted from EV lysates in a polymer-coated tube. Subsequently, isothermal SCOPE reactions occur in that same tube, producing analytical results within 30 min. Starting from sample loading, the entire process is completed within 1 h, facilitating same-day preclinical and clinical molecular diagnostics. Moreover, the assay uses standard laboratory equipment (for example, size exclusion column for EV isolation and thermal cycler) and requires only small volumes of samples (for example, EV isolates from <100 µl of plasma).

SCOPE instrumentation

We implemented compact modules to facilitate the assay process and enhance end-user friendliness. The first module was a polymer-coated tube for nucleic acid extraction (Fig. 2a). Specifically, we prepared PCR tubes coated with a charge-shifting polymer, poly[(2-dimethylamino) ethyl acrylate] (pDMAEA), through initiated chemical vapor deposition (see Methods for details; Supplementary Fig. 3a). Upon addition of an aqueous sample, the polymer inside the tube dissolved and became positively charged, allowing for electrostatic interactions between cationic polymers and negatively charged nucleic acids. This reaction resulted in the formation of polyplexes, which could be collected via centrifugation (5 min, 123g). Subsequently, the SCOPE mixture was added to the precipitates, and the temperature was raised to 40 °C. Under this condition, the polymer underwent self-catalyzed hydrolysis of the ester groups located in its side chains. This process induced negative charges on the polymer, resulting in the disassembly of polyplexes and the release of nucleic acids. The SCOPE reaction then continued within the same tube. We observed no significant difference ($P > 0.05$, unpaired two-sided *t*-test) in SCOPE signals when using pDMAEA-coated tubes compared to standard PCR tubes (Supplementary Fig. 3b). In addition, self-catalyzed hydrolysis of pDMAEA was confirmed through nuclear magnetic resonance (NMR) spectroscopy (Supplementary Fig. 4).

Figure 2b compares the RNA extraction efficiency between pDMAEA tubes and standard silica-based columns (Methods)²⁸. The pDMAEA approach achieved similar extraction yields as the column devices, with an average extraction yield of 96% across different input RNA amounts (3, 30, 60 and 100 ng). Using pDMAEA tubes, however, streamlined the overall assay workflow. It facilitated rapid nucleic acid extraction (10 min) and enabled the execution of SCOPE reactions within a single closed-tube format, eliminating the need for sample transfer in between steps.

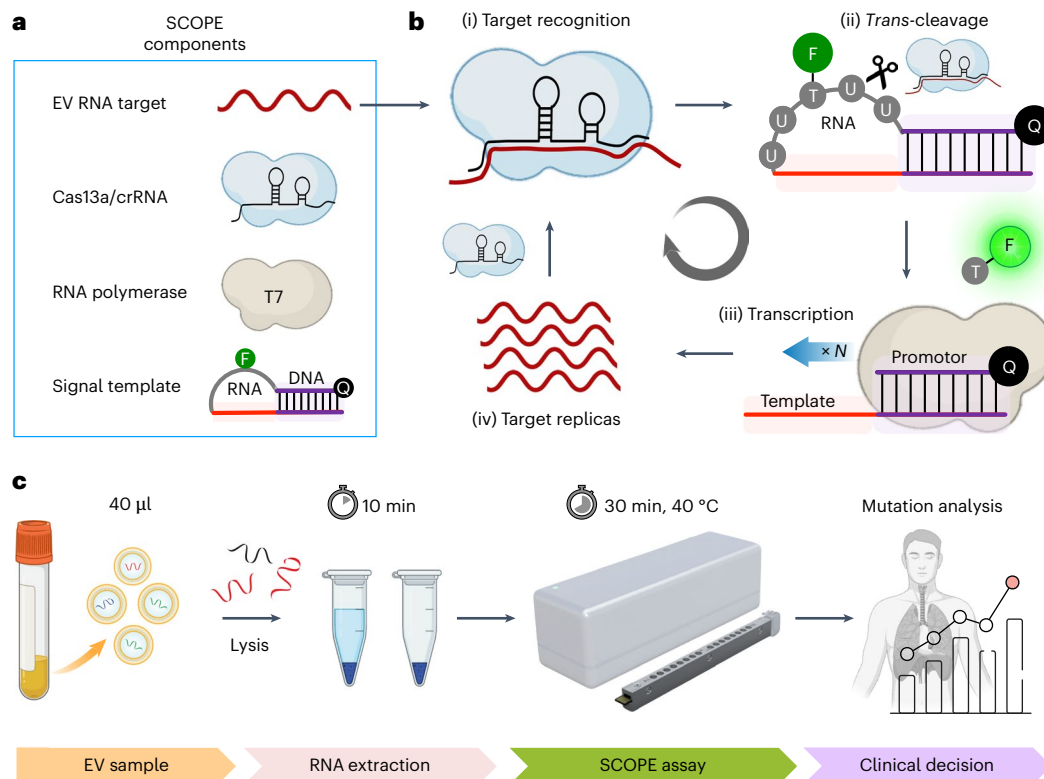


Fig. 1 | SCOPE. a, SCOPE mechanism. The assay couples Cas13a/crRNA and polymerase reactions through a signal template (bottom). The signal template contains an RNA segment with a quenched fluorescent dye (F, fluorescent dye; Q, quencher). **b**, Assay flow. (i) Cas13a/crRNA first recognizes the EV RNA target, which activates its ribonuclease function. (ii) The activated enzyme cuts the RNA segment in the signal template, generating a fluorescent signal and exposing the DNA template. (iii) T7 RNA polymerase binds to the promoter region of the

DNA template and synthesizes target RNA replicas. (iv) The amplified target replicas bind to free Cas13a/crRNA, enforcing the overall assay process. Note that the initial CRISPR-mediated target recognition is necessary to unlock RNA replication and signal generation. **c**, SCOPE workflow for onsite diagnostics. EVs are isolated from clinical samples and lysed. EV lysates are then placed in surface-treated tubes for RNA extraction (10 min). Subsequently, the SCOPE reaction is performed in a compact, portable device (30 min). The assay produces molecular information within 1 h, enabling same-day clinical decisions.

The second assay module was a compact device for parallel SCOPE measurements (Fig. 2c and Supplementary Fig. 5). It integrated a tray-type heating block, a fluorescent optical detector and a line scanner. The heating block could hold up to 16 PCR tubes and docked to the main frame for an electrical connection. The system then generated pre-programmed temperature profiles (for example, isothermal and thermal cycling). Once the reaction was complete, the optical detector measured fluorescent signals (Fig. 2d). The detector comprised two independent fluorescent excitation/detection headers, with each header performing independent measurements. Two identical headers could be installed for redundant one-color measurements, or two different modules could be used for two distinct fluorescent dyes. The optical detector executed a linear motion to scan all 16 samples. System operation was controlled through a graphical user interface on an external terminal device.

The developed SCOPE system demonstrated high uniformity and reproducibility in parallel detection. The sample holder heated all 16 samples to 40 °C, with well-to-well temperature variations of less than 0.5 °C (Supplementary Fig. 5). We also observed highly uniform SCOPE signals for samples containing the same amount of mRNA, regardless of their location in the heating block (Fig. 2e).

Validating SCOPE concept

We systematically examined SCOPE assay kinetics through analytical modeling (see Supplementary Note for details). SCOPE couples two distinct catalytic reactions through the signal template (Fig. 3a): (1) target-bound Cas13a/crRNA turns on the fluorescent signal by

cleaving the RNA segment in the signal template and (2) RNA polymerase replicates target mRNAs using the DNA sequence in the signal template. When executed separately, each type of catalytic activity generated the end product (that is, fluorescent signal or mRNA replica) that increased linearly with time (Fig. 3b, left); these observations matched the zeroth-order nature of Cas13a and polymerase reactions in our assay conditions (Supplementary Note)^{29,30}. In contrast, coupling these two processes approximated a first-order rate kinetics (Supplementary Note), speeding up the overall reaction. Indeed, SCOPE signals rose exponentially and reached a plateau within 30 min (Fig. 3b, right).

Notably, off-target RNA amplification was suppressed. For RNA amplification to proceed, SCOPE first required Cas13a/crRNA to recognize its initial RNA target. The activated Cas enzyme then degraded the RNA segment in the signal template, which triggered downstream polymerase activities (Supplementary Fig. 6). Conversely, the polymerase failed to replicate RNA targets when signal templates had an intact RNA segment (Supplementary Fig. 7); the overall loop structure presumably hindered the polymerase from docking the promoter region of the signal template.

We further optimized the SCOPE template design and reaction conditions to maximize its signal intensity (Supplementary Fig. 8). The optimal assay was then validated by controlling the reactant composition (Fig. 3c). The analytical signal was highest when all assay components (for example, target RNA, Cas13a, crRNA and T7 RNA polymerase) were present. The signal intensity decreased when RNA polymerase was absent; therefore, no additional targets were synthesized.

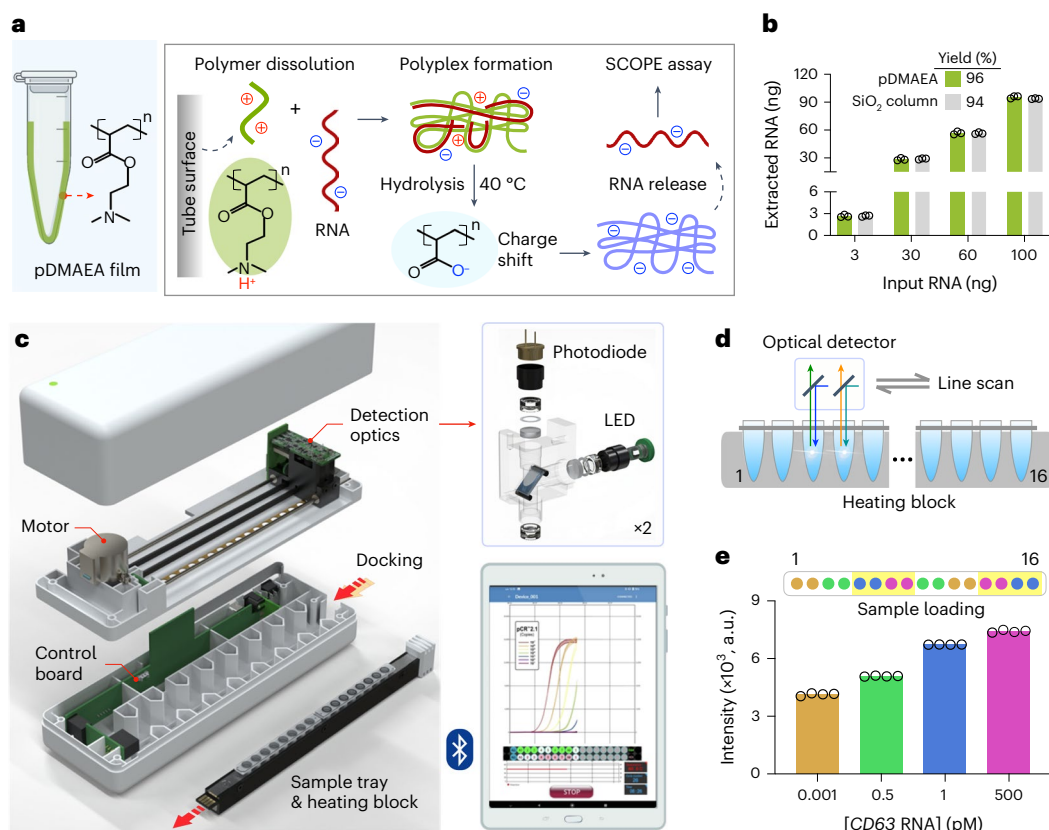


Fig. 2 | SCOPE device engineering. **a**, Single-tube system for SCOPE assay. We used PCR tubes coated with pDMAEA polymer. With EV lysates added, the dissolved polymer was positively charged initially, forming polyplexes with negatively charged nucleic acids. After brief centrifugation to collect polyplexes, we added SCOPE reagents. The polymer underwent charge-shifting hydrolysis to release nucleic acids, and the SCOPE reaction continued in the same tube. **b**, Comparison of RNA extraction yield between pDMAEA tubes and standard silica-based columns. The pDMAEA method demonstrated similar extraction yields as the column devices, with an average extraction yield of 96% across different input RNA amounts (3, 30, 60 and 100 ng). Extracting nucleic acids with pDMAEA tubes was fast (10 min) and compatible with SCOPE reactions. Data are shown as mean $\pm$ s.d. from technical triplicates. **c**, Integrated device for onsite SCOPE tests. This compact device accommodated 16 samples in a

tray-like sample holder. The tray was wrapped with a flexible heater and docked to the main frame that regulated the heater. Fluorescent signals were detected by an optical module (top right). The system was connected to a tablet via wireless communication. A user-friendly graphical interface (bottom right) in a tablet provided intuitive windows for system operation and data analysis. **d**, Upon completion of the reaction, the optical module quantified fluorescent signals emitted from the samples. A single module linearly scanned the sample tray, simplifying the system design. **e**, The SCOPE device was used to process samples containing different amounts of synthetic *CD63* RNA. The SCOPE results were highly consistent for a given RNA concentration, regardless of the sample location in the heating block. The coefficient of variation was less than 1.5%. The bar represents a mean fluorescent signal from four different sites in the heating block. Color codes matched the sample location with RNA concentration.

The signals were negligible in those cases wherein initial target recognition by Cas13a/crRNA was missing.

SCOPE assay characterization

SCOPE achieved high sensitivity through its dual amplification scheme (fluorescent signal and mRNA target). In RNA titration experiments, SCOPE's limit of detection (LOD) reached sub-attomolar ranges, which was more than 10^9 -fold lower than PCR with reverse transcription's (RT-PCR) LOD (Fig. 4a for *KRAS*^{G12D}; Supplementary Fig. 9 for *CD63*). Moreover, SCOPE displayed a broader dynamic range than RT-PCR. We also observed an enhancement in sensitivity when analyzing EVs as mRNA sources (Fig. 4b); the LOD by SCOPE was about 10^5 EVs per milliliter, which was more than 10^3 -fold lower than the LOD by RT-PCR.

SCOPE also exhibited superb sequence specificity, allowing for precise differentiation of point mutations. We chose *KRAS*, the most frequently mutated oncogene, as a representative target and prepared SCOPE mRNA probes for wild-type *KRAS* (*KRAS*^{WT}) and its mutated subtypes, *KRAS*^{G12C}, *KRAS*^{G12D}, *KRAS*^{G12S} and *KRAS*^{G12V} (Supplementary Table 1). SCOPE consistently demonstrated high signal-to-background ratios between on-target and off-target samples (Fig. 4c and Supplementary Fig. 10); for 1 nM target concentrations,

the ratio was greater than 30 across different *KRAS* targets. SCOPE's high selectivity enabled reliable analyses of samples with low variant allele fractions (VAFs). For instance, we could detect *KRAS* mutant alleles down to 0.01% VAF (Fig. 4d), a performance on par with digital PCR and sequencing^{31,32}. In contrast, RT-PCR assays displayed poor specificity due to off-target crosstalk, even when a commercial *KRAS* kit was used (Supplementary Fig. 11).

Early detection of lung cancer in an animal model

We hypothesized that SCOPE's high sensitivity could facilitate EV-based early cancer detection. To test the hypothesis, we employed a genetically engineered mouse model of lung adenocarcinoma (KP model), characterized by Cre-induced *Kras*^{G12D/+} *p53*^{-/-} mutation³³. This autochthonous model allowed us to regulate tumor initiation and collect blood samples at different stages of tumor development (Fig. 5a), an approach that effectively replicated the challenge of detecting EVs originating from early lesions.

We first validated whether EV mRNA profiling can infer cellular genotypes. For this validation, we used cell lines with different *KRAS* genetic backgrounds, specifically H2228 (*KRAS*^{WT/WT}), A549 (*KRAS*^{G12S/G12S}) and a well-characterized cell line derived from the KP

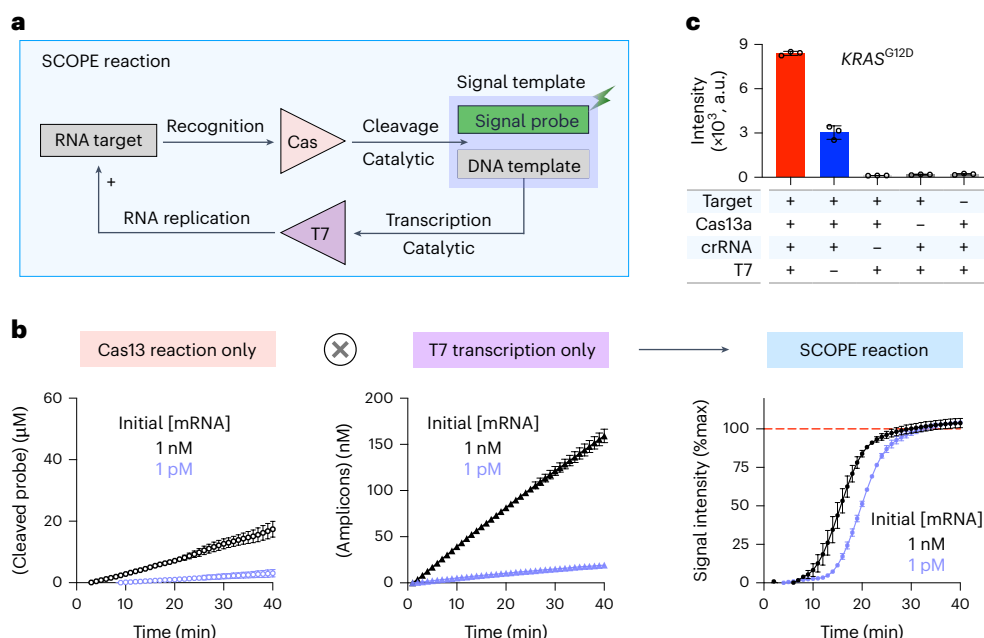


Fig. 3 | SCOPE assay kinetics. **a**, SCOPE integrates two enzymatic reactions via the signal template: (1) Cas13a/crRNA generates fluorescence by degrading the RNA segment in the signal template and (2) T7 polymerase replicates RNA targets. SCOPE reactions commence upon initial recognition of mRNA targets by Cas13a/crRNA. **b**, When performed independently, Cas13a/crRNA and T7 reactions exhibited a linear increase in end product over time (left and middle panels). Coupling these two reactions enabled SCOPE to emulate first-order rate kinetics. The analytical signals amplified exponentially, reaching a steady

state within 30 min (right panel). Data are shown as mean $\pm$ s.d. from technical triplicates. **c**, Assay validation. Signal intensity was maximized when all SCOPE assay components were present (red bar). Removing T7 polymerase diminished the signal due to the absence of target RNA amplification (blue bar). Assay conditions that hindered the initial mRNA recognition produced negligible signals (gray bars). The initial target RNA (KRAS^{G12D}) concentration was fixed at 1 nM. Data are shown as mean $\pm$ s.d. from technical triplicates.

mouse model (KP1.9;Kras^{G12D/WT})³⁴. EVs were collected from cell culture media and subjected to the SCOPE KRAS mRNA assay (Methods). We observed that EV profiling results consistently matched cellular KRAS status (Fig. 5b). EVs from KP1.9 cells exhibited high SCOPE signals for Kras^{WT} and Kras^{G12D}, reflecting the cell line's Kras heterozygosity. In contrast, EVs from homozygous cell lines were exclusively positive for one of the KRAS targets.

We next applied SCOPE to analyze EVs in mouse plasma samples. We prepared two cohorts using KP mice with identical genetic backgrounds (Kras^{G12D/+}). The tumor-bearing group received an intranasal inoculation of Cre adenovirus to induce oncogene expression and initiate tumor growth, whereas the non-cancer control group did not undergo the procedure. Immunohistochemistry (IHC) performed on lung tissue specimens confirmed tumor growth in the Cre-treated mice (Fig. 5c). After oncogene induction, multifocal disease developed that heterogeneously progressed from atypical adenomatous hyperplasias and small adenomas into larger adenomas and eventually invasive and disseminated adenocarcinoma³⁵. Notably, Kras^{G12D} protein expression in lung epithelium was observed within 1 week after Cre induction, followed by tumor formation in the subsequent weeks (Fig. 5c).

For both cancer ($n = 10$) and control ($n = 3$) cohorts, we collected longitudinal blood samples and analyzed plasma EVs for Kras^{G12D} mRNA; such serial EV profiling was feasible because SCOPE required only 40 μl of plasma samples. For tumor-bearing animals, we also obtained computerized tomography (CT) scans on a weekly basis. Figure 5d and Supplementary Fig. 12 display Kras^{G12D} SCOPE signals tracked over 8 weeks. In the non-cancer mice, the signal remained at background values throughout the duration. Conversely, in tumor-bearing mice, the signal increased from the background once oncogene expression was induced and reached saturation at about 5 weeks of tumor growth. CT analysis performed on the same animals demonstrated tumor growth

over time, although the lesions became detectable approximately 2 weeks after tumor induction (Supplementary Fig. 13).

We further estimated mRNA copy numbers (Fig. 5e and Supplementary Fig. 14) by referencing the SCOPE results to the titration curve (Fig. 4a). In cases without tumors, we observed a consistent baseline (copy number <1). However, the copy number became significantly different from the baseline as early as 1 week after oncogene induction ($P = 0.042$, unpaired t -test). These findings support SCOPE's potential to detect early tumors, minimal residual disease and/or recurrences, especially in cases involving known mutations.

Monitoring patients with colorectal cancer during clinical care

RAS is frequently mutated in CRC, with about 40% of CRC cases showing these mutations, primarily in codon 12 (65%)^{13,36}. Assessing and monitoring KRAS mutations remain crucial for prognosis evaluation and treatment decisions. SCOPE could streamline this process by enabling minimally invasive, accessible and highly accurate KRAS profiling. To test this approach, we applied SCOPE to analyze EVs for KRAS^{WT} and major mutations in codon 12 (KRAS^{G12C}, KRAS^{G12S}, KRAS^{G12V}). The KRAS SCOPE probes were first validated with EVs derived from a panel of CRC cell lines, each representing a different KRAS codon 12 mutation status (Supplementary Fig. 15).

After probe validation, we proceeded to a clinical study as outlined in Fig. 6a (see Supplementary Table 2 for the patient information). We collected pre-surgical plasma samples from patients with CRC who had not undergone any treatment. Concurrently, tissue samples were acquired and analyzed for KRAS mutations. For those patients with codon 12 KRAS mutations confirmed in tissue samples, we further examined longitudinal plasma samples collected after surgery and during standard adjuvant chemotherapy (1 year). For EV isolation, we processed plasma samples through size exclusion columns (Methods).

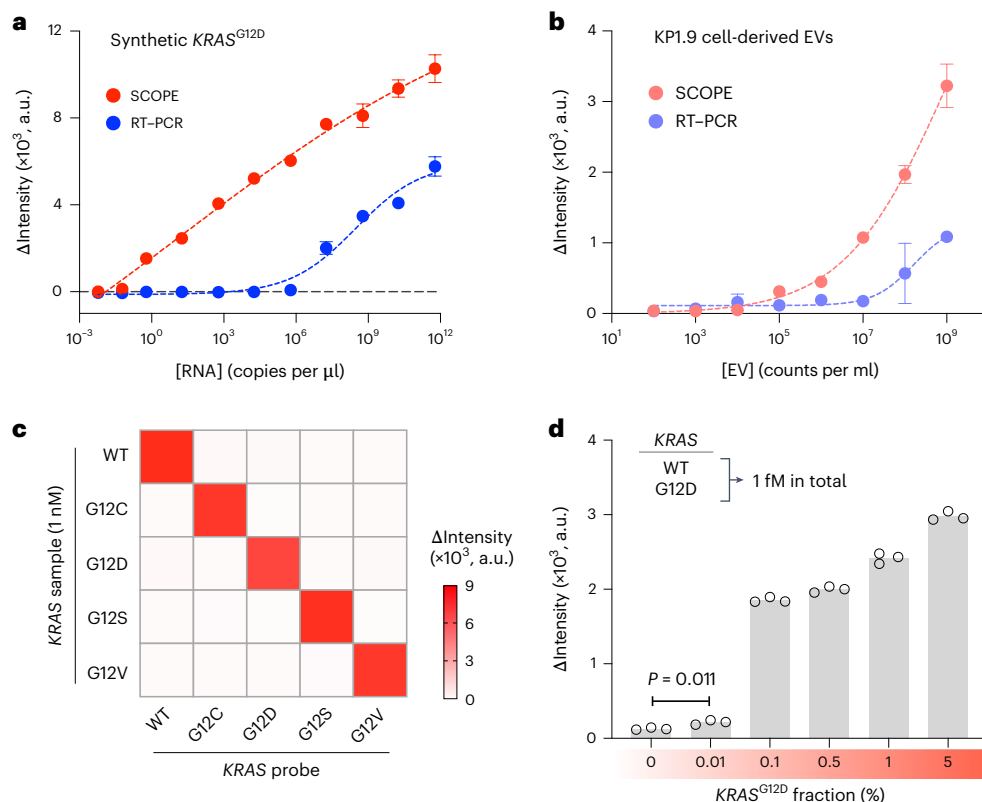


Fig. 4 | SCOPE assay characterization. **a**, SCOPE (30 min) and conventional RT-PCR (80 min) were employed to analyze samples containing varying concentrations of synthetic *KRAS*^{G12D} RNA. SCOPE exhibited an LOD of 8.5×10^{-3} copies per microliter (14.2 zM), outperforming RT-PCR (LOD of 1.1×10^7 copies per microliter, 18.6 pM). Data are displayed as mean $\pm$ s.d. from technical triplicates. Some error bars are too small to be visible. The net intensity (Δ Intensity) was obtained by removing background fluorescent signals (no-target controls). **b**, EV samples were prepared from KP1.9 cell culture and analyzed for *Kras*^{G12D} mRNA. SCOPE's LOD was approximately 10^5 EVs per milliliter, whereas RT-PCR's LOD was approximately 10^8 EVs per milliliter.

In addition, SCOPE displayed a broader dynamic range than RT-PCR. Data are shown as mean $\pm$ s.d. from technical triplicates. **c**, The specificity of SCOPE probes targeting *KRAS*^{WT} and its mutated subtypes (G12C, G12D, G12S and G12V) was evaluated. SCOPE displayed high signal contrast between on-target and off-target samples. At a target concentration of 1 nM, the contrast ratio exceeded 30. The heatmap shows mean values from technical triplicate measurements. **d**, A mixture of *KRAS*^{WT} and *KRAS*^{G12D} synthetic RNA samples was prepared, varying the VAFs. SCOPE detected *KRAS*^{G12D} down to 0.01% VAF, with a signal significantly different from the background ($P = 0.011$, unpaired, two-sided t -test). The bars represent mean values from technical triplicate measurements.

The collected EV fraction showed positive signals for established EV markers (CD63 and Alix) while lacking a non-EV marker (histone H2B), indicating EV enrichment in our samples (Supplementary Fig. 16a)³⁷. Furthermore, when we applied the SCOPE assay to both EV and non-EV fractions, only the EV fraction reported high SCOPE signals for mRNA targets (Supplementary Fig. 16b). Treating the EV isolates with RNase did not elicit significant changes to the SCOPE signal (Supplementary Fig. 16c). Collectively, these findings strongly suggested that EV-associated mRNA was the primary source of signals detected by the SCOPE assay.

Figure 6b shows the initial EV profiling results of samples from pre-surgical patients with CRC ($n = 107$) and non-cancer controls ($n = 15$) (see Supplementary Fig. 17 for the complete heatmap). The SCOPE assay identified a subgroup of patients with CRC ($n = 17$) whose mutation types matched their tissue *KRAS* status. This finding was further corroborated through tissue staining (IHC), which detected mutated *KRAS* protein in tumor tissues (Fig. 6c and Supplementary Fig. 18). These observations underlined the rationale for assessing tumor *KRAS* status through EV profiling.

To evaluate the prognostic potential of detecting *KRAS* mutation via SCOPE, we used the ratio $R_{KRAS} = KRAS^{MT}/KRAS^{WT}$ between *KRAS* mutation (*KRAS*^{MT}) and *KRAS*^{WT} signals. We then set a threshold for this ratio to determine whether a sample is positive for *KRAS* mutation. The data from EV samples without *KRAS* mutation (that is, non-cancer controls and patients with *KRAS*^{WT} CRC) were pooled and fitted to a

normal distribution (Supplementary Fig. 19). We then set the threshold for *KRAS* mutation negativity as the 99th percentile of this distribution ($R_{KRAS} = 5.5\%$). Applying this threshold, we could effectively categorize samples by their *KRAS* mutation status (Fig. 6d). In samples with *KRAS* mutation, only the R_{KRAS} value of the mutated gene exceeded the threshold. In contrast, the values of non-mutated genes remained below it.

We further monitored the R_{KRAS} of the 17 patients with *KRAS* mutations. Fourteen patients were eventually deemed non-recurrent, with the remaining three considered recurrent; these were confirmed by surgical resection or radiological detection of lesions. Figure 6e summarizes the changes in R_{KRAS} values of these patients (see Supplementary Fig. 20 for individual patient results) as they received standard clinical care, including curative surgery and chemotherapy. For all 17 patients, R_{KRAS} initially decreased after surgery, presumably due to the reduction in CRC-derived EVs in circulation after tumor resection. However, a noticeable divergence emerged over time. Among the non-recurrent patients (Fig. 6e, left), R_{KRAS} continued to decrease and eventually fell below the non-*KRAS*^{MT} threshold, aligning with favorable clinical outcomes (tumor-free status). In contrast, R_{KRAS} values rebounded in recurrent patients and ultimately returned to their pre-surgery levels (Fig. 6e, right).

R_{KRAS} values for mutated mRNA were higher than the threshold (5.5%) in the days soon after surgery. We estimated EV clearing by modeling the R_{KRAS} trend in non-recurrent patients (Fig. 6f). The R_{KRAS} values decreased over time according to a power law ($R_{KRAS} \sim d^{0.3}$), where d is

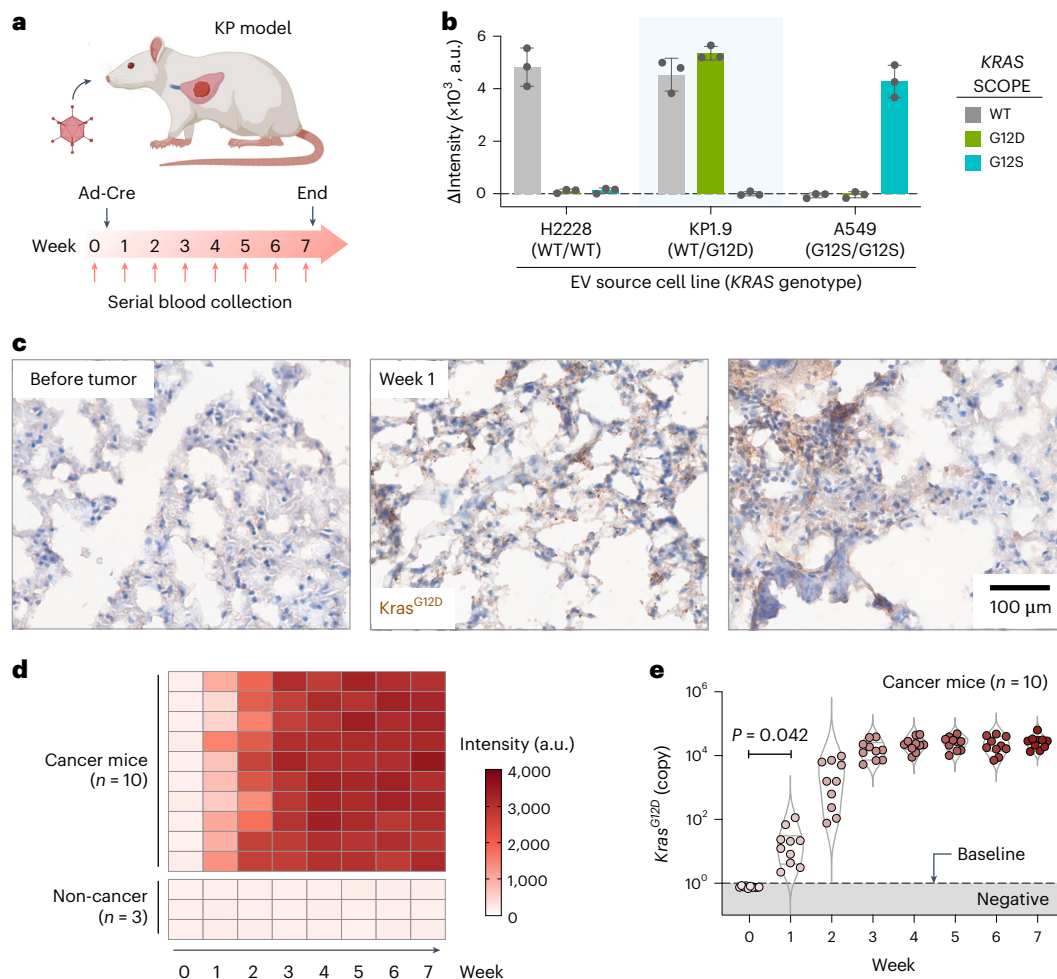


Fig. 5 | Early cancer detection with SCOPE. **a**, Experimental design. A genetically engineered mouse model was used to mimic the development of non-small cell lung cancer. The animal had a Cre-activatable *Kras*^{LSL-G12D/+}; *Trp53*^{fllox/fllox} genetic background (KP model). Tumor growth was initiated through intranasal administration of adenoviruses expressing Cre recombinase (Ad-Cre). Serial blood samples were collected from the animals before tumor induction and at weekly intervals after that. **b**, In vitro validation of SCOPE for *KRAS* genotyping. Cell lines with distinct *KRAS* genetic profiles were used: H2228 (WT/WT), KP1.9 (G12D/WT) and A549 (G12S/G12S). EVs were isolated from cell culture media. SCOPE *KRAS* assays on these EV samples yielded results consistent with the cellular *KRAS* status. Data are presented as mean ± s.d. from biological triplicates. **c**, Tumor development in the Cre-treated mice was corroborated via IHC on lung tissue specimens. Lesions positive for *Kras*^{G12D} protein were detectable 1 week after tumor induction, and the lesions enlarged in subsequent weeks. Staining

was performed on two different samples, and representative images are shown. **d**, SCOPE analyses on EV *Kras*^{G12D} mRNA during tumor development. Serial blood samples were collected over 8 weeks from the same individual mice. In the non-cancer cohort ($n = 3$), the SCOPE signal remained at background values throughout the duration. In the tumor-bearing cohort ($n = 10$), the SCOPE signal increased from background values 1 week after tumor induction and reached saturation approximately 3 weeks after the onset of tumor growth. The heatmap shows mean values from technical triplicate measurements. **e**, Estimation of *Kras*^{G12D} mRNA copy numbers. SCOPE results in **d** were converted based on the titration curve (Fig. 4a). In the tumor-bearing cohort, the copy number was below the baseline (<1) initially and then significantly increased ($P = 0.042$, paired two-sided t -test) 1 week after tumor induction. Each data point is a mean value from technical triplicate measurements.

the number of days after surgery. The model suggested that R_{KRAS} values would fall below the mutation threshold about 10 d after surgery, which underscored the potential advantage of delaying EV analysis for a more accurate assessment of minimal residual disease and post-surgical prognosis. Furthermore, these analyses present an opportunity to develop an EV-based prognostic metric to guide the selection and duration of adjuvant treatment.

Stratifying patients with glioma

We used SCOPE to detect a panel of genetic alterations in patients with glioma, specifically IDH1 mutation, EGFR amplification and EGFRvIII deletion. Obtaining this information can be crucial to classifying gliomas and guiding effective treatment based on their molecular traits³⁸. For instance, patients with IDH1-WT and EGFR amplification (classified as glioblastoma IDH1-wt) are linked to poor prognoses and resistance

to radiotherapy³⁹. On the other hand, patients with the IDH1 mutation (classified as astrocytoma or oligodendroglioma depending on other markers, including 1p19p co-deletion) exhibit prolonged overall survival and are more likely to respond to temozolomide treatment^{40,41}. The analysis of EVs in peripheral blood holds particular promise for gliomas, considering the risks associated with complications and morbidity from brain tissue biopsies.

To investigate EV-based glioma typing, we analyzed EVs in plasma samples taken before surgery. Results were compared to those derived from corresponding tissue samples collected during surgical removal of tumors (Extended Data Fig. 1a). We first designed and validated a new set of SCOPE probes for mRNA targets (Extended Data Fig. 1b and Supplementary Fig. 21): WT *IDH1* (*IDH1*^{WT}), its point mutation (*IDH1*^{R132H}) most frequently found in gliomas, WT *EGFR* (*EGFR*^{WT}) and *EGFRvIII*. The designed probes were highly specific (Extended Data Fig. 1c) and

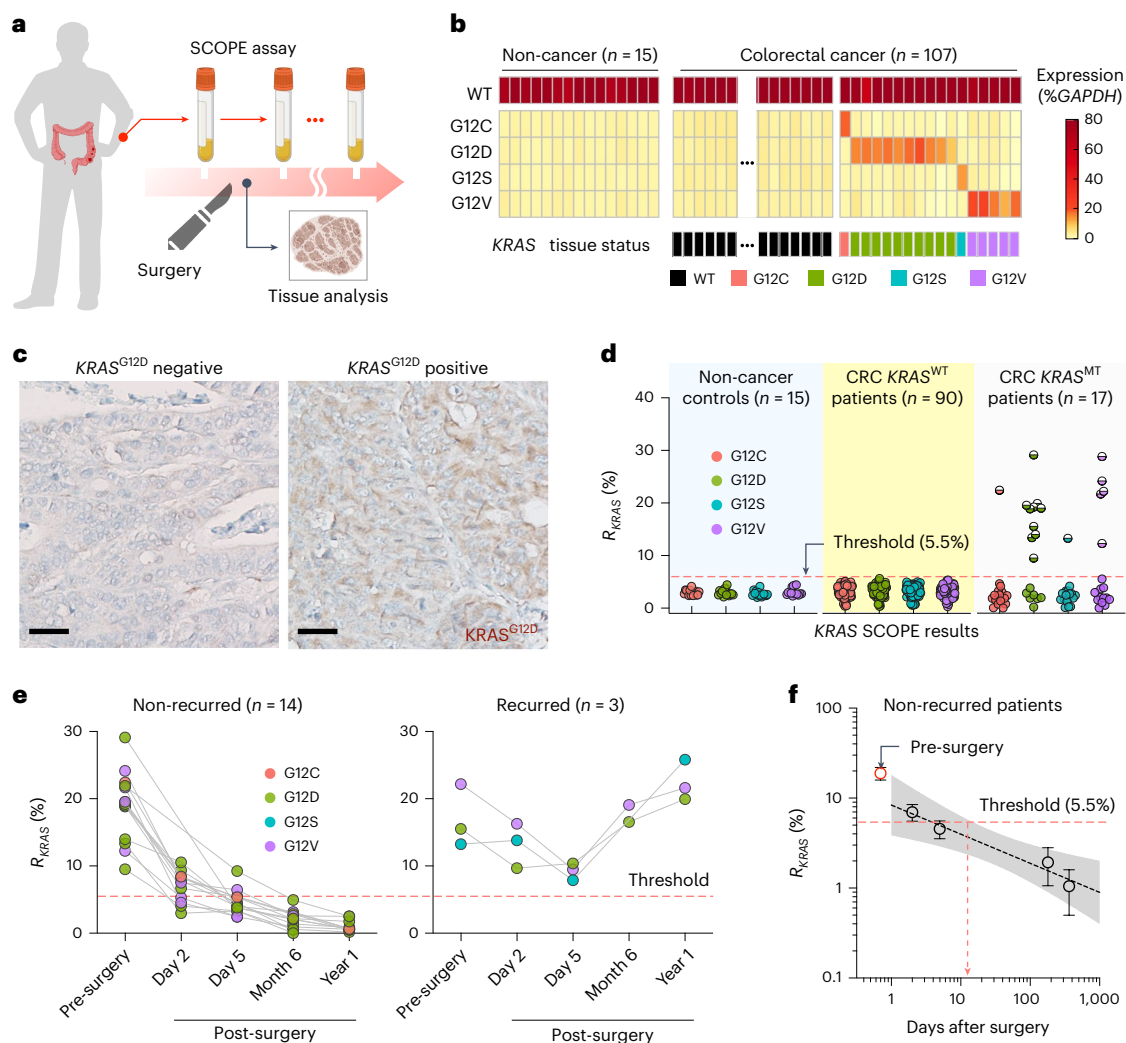


Fig. 6 | Monitoring patients with CRC. **a**, Plasma samples were collected from patients with CRC before and after surgery as well as during standard care. Tissue samples were obtained during surgery and analyzed for the *KRAS* gene. **b**, Plasma EVs were analyzed via SCOPE for a panel of *KRAS* mRNA mutations. Samples were from non-CRC controls ($n = 15$) and pre-surgery patients with CRC ($n = 107$). SCOPE results reflected *KRAS* mutation status from tumor tissue analysis. *KRAS* signals from EVs were normalized against that of *GAPDH*. The heatmap shows mean values from technical triplicates. **c**, Images of tissue staining (IHC). For a patient positive with *KRAS*^{G12D} EV mRNA, the tumor tissue was also positively stained for *KRAS*^{G12D} protein. The staining experiments were performed on two different tissues, and representative images are shown. Scale bars, 20 μ m. **d**, The results from **b** were converted into the ratio ($R_{KRAS} = KRAS^{MT}/KRAS^{WT}$) of SCOPE signals between specific *KRAS*^{MT} and *KRAS*^{WT}. The threshold for *KRAS*^{MT} positivity was established using profiling data from non-CRC controls and patients with *KRAS*^{WT} CRC. For each specific *KRAS* mutation, the R_{KRAS} values of patients with

CRC harboring that particular mutation exceeded those of non-CRC controls and patients with *KRAS*^{WT} CRC. Half-filled circles ($\odot$) indicate patients whose *KRAS* mutation type is concordant between SCOPE and clinical tissue analyses. Each data point is a mean value from technical triplicates. **e**, Seventeen patients with *KRAS*^{MT} CRC were followed up for 1 year. For all patients, R_{KRAS} values immediately after surgery were lower than their initial values. Among the non-recurrent patients ($n = 14$), R_{KRAS} decreased below the threshold (5.5%). In the recurrent patients ($n = 3$), R_{KRAS} values eventually returned to pre-surgery levels. Each data point is a mean value from technical triplicates. **f**, In the non-recurrent patients with CRC, the R_{KRAS} values would fall below the threshold approximately 10 d after surgery. The black dashed line is the optimal fit for the post-surgery data, $R_{KRAS} = d^{0.3}$, where d is the number of days after surgery. The shaded region represents the 95% confidence interval. Data are shown as mean $\pm$ s.d. from 14 patients.

enabled SCOPE to genotype different GBM cell lines through EV profiling (Extended Data Fig. 1d).

Next, we analyzed plasma samples from 15 non-cancer controls and 60 patients with glioma with different tumor subtypes (see Supplementary Table 3 for patient information). We applied SCOPE to profile EVs for *GAPDH*, *CD63*, *IDH1*^{WT}, *IDH1*^{R132H}, *EGFR*^{WT} and *EGFRvIII* mRNA. Given that both *EGFR*^{WT} and *EGFRvIII* were designated as target markers, we used the *GAPDH* gene as a reference for normalization purposes^{14,42}. Specifically, we scaled signals from other markers against the *GAPDH* signal rather than estimating VAFs. We also molecularly assessed and classified patients' tumor specimens (Methods). The SCOPE results demonstrated the potential of robust and highly sensitive glioma

typing through EV analysis (Extended Data Fig. 1e and Supplementary Fig. 22). We observed differential expression patterns of *EGFR*^{WT}, *IDH1*^{R132H} and *EGFRvIII* across different glioma subtypes. For example, *EGFR*^{WT} expression in EVs was higher in patients with GBM with *EGFR* amplification or *EGFRvIII* mutation than in patients with tumors harboring *IDH1*^{R132H} mutation. *IDH1*^{R132H} and *EGFRvIII* mRNA were only detected in the EVs from patients with matching mutations in tumor tissue.

Discussion

EVs are a promising source of cell-free RNA for liquid biopsy applications both in preclinical and clinical contexts. Nucleic acids within EVs are shielded from degradation and are more abundant than

ctDNA, which renders EV RNA robust analytical targets for molecular diagnostics^{14,43}. Among the various EV RNA species, microRNAs are highly abundant and, thus, more commonly studied^{18,19}. In contrast, due to its lower concentration, detecting EV mRNA often requires large sample volumes and advanced instrumentation. The SCOPE technology can overcome these limitations by improving target amplification and signal generation. (1) SCOPE exhibits specificity down to a single nucleotide. This precision is achieved through an innovative concept where RNA replication is initiated solely upon recognition of target RNA by Cas13a. (2) SCOPE's concurrent amplification of RNA targets and fluorescent signals leads to analytical sensitivity far surpassing other isothermal amplification methods (see Supplementary Table 4 for comparison)⁴⁴. (3) SCOPE produces these results within a short time-frame (30 min) through a one-pot assay that can be conducted using readily available laboratory tools. These advantages would facilitate SCOPE's integration into routine preclinical and clinical applications across academic and industrial laboratories.

A key innovation of SCOPE lies in its signal template, a hybrid design that combines a cleavable, fluorescent RNA probe with a DNA template. This structure synergistically couples two enzymatic activities (Cas13a collateral cleavage and T7 polymerase reaction) and establishes a self-amplifying cycle to accelerate signal growth. Notably, this reaction is conditional—Cas13a first needs to recognize the RNA target to unlock the subsequent SCOPE reaction. Our rationale for this design was supported by kinetic modeling and experimental data. When Cas13a and T7 reactions were executed individually, their assay products increased linearly over time, consistent with zeroth-order kinetics. Conversely, SCOPE signals displayed exponential growth and rapidly reached saturation, closely resembling first-order rate kinetics. However, we acknowledge that the reaction scheme may be susceptible to false positives. A potential contributing factor is the degradation of the fluorescent RNA segment in the signal template. This segment, designed for specific cleavage by the activated Cas13a, can be degraded by other RNases present in samples. In the current work, we mitigated such an effect by including a broad-spectrum RNase inhibitor in the SCOPE mixture. We also analyzed technical replicates of each sample to guard against false positives. In future work, we may consider chemically modifying the RNA segment, for instance, with phosphorothioate to enhance nuclease resistance⁴⁵. Further investigations would be necessary to evaluate Cas13a *trans*-cleavage efficacy on these modified targets.

We validated SCOPE's performance through systematic pre-clinical and clinical studies. SCOPE's LOD extends into the sub-attomolar ranges. Equally impressive was SCOPE's high selectivity, enabling the precise identification of point mutations. In a noteworthy example, SCOPE effectively differentiated *KRAS*^{WT} mRNA from its mutated subtypes (G12C, G12D, G12S and G12V) with minimal crosstalk, surpassing the performance of a commercial *KRAS* mutation assay. Furthermore, SCOPE detected *KRAS* mutant alleles at levels as low as 0.01% VAF, a sensitivity similar to high-end techniques such as digital PCR and BEAMing PCR⁴³. The SCOPE assay achieved this sensitivity without sample partitioning and remained compatible with conventional thermal cyclers, which would enhance its accessibility to a wider research community.

SCOPE demonstrated potential for various clinical applications, including early tumor detection, recurrence monitoring, residual disease assessment and tumor subtyping. The assays were fast (<30 min for signal generation) and cost-effective (<\$4 per marker). These findings illustrate SCOPE's versatility and affordability as a diagnostic tool to expedite clinical decision-making (same-day turnaround), track cancer recurrence/minimal residual disease, gauge responses to therapies (both conventional and experimental under trial auspices) and achieve timely pre-clinical readouts to support drug development efforts. The current results are pertinent to size exclusion chromatography (SEC)-processed plasma samples, a commonly used approach across EV studies⁴⁶. We employed SEC throughout the study for consistent

sample handling, but this method may co-isolate additional extracellular RNA carriers, such as lipoproteins and platelet-derived EVs, that could contribute to the overall SCOPE signals. To further elucidate specific mRNA carriers, we may consider a single-vesicle imaging study; this approach will involve fluorescent labeling of vesicle surface proteins (for example, CD9 for EVs and CD31 for platelets) to identify their origins^{9,47} while simultaneously labeling target mRNA with molecular beacons. Subsequent imaging of these vesicles would reveal the origin of mRNA-carrying vesicles⁴⁸. We could also conduct studies that evaluate SCOPE's applicability with other sample types and pre-processing methods (see Supplementary Fig. 23 for examples) to examine SCOPE's robust performance across diverse use cases.

By employing the SCOPE technique, researchers could expand their knowledge of EV characteristics. In this work, we detected circulating EVs harboring genetic mutations that mirrored tumor tissues, thereby reaffirming EVs' potential as biomarkers. Longitudinal monitoring yielded further insights into EV dynamics. (1) Mutation targets in EVs progressively increased during tumor initiation and growth. (2) EV mutational loads declined from pre-surgery values after curative surgery in patients with CRC. This decline persisted in patients responsive to chemotherapy and receded to the background value. These observations contrasted with non-responsive patients with CRC whose EV mutational loads stayed above the background value. These results underscore the potential utility of EVs for tumor monitoring, akin to the capabilities demonstrated with ctDNA⁴⁹. We also note unexpected observations that warrant further investigation. Within 1 week of CRC surgery, EV mutational loads may remain elevated, potentially due to residual microscopic tumors or the release of tumor-derived materials during surgery. In our pilot study, this uncertainty resolved approximately 10 d after surgery, a potentially optimal window to begin EV analysis to inform post-surgery prognoses and shape the type and duration of adjuvant interventions. A valuable next step would be to broaden SCOPE to include diverse tumor types and assess its applicability within multimodal treatment contexts. This expansion could establish cancer-specific timeframes for optimal EV analysis, furthering its utility in treatment monitoring and prognostication.

Another potent strategy entails analyzing both EV mRNA and ctDNA in plasma to obtain maximal molecular information. EV mRNA possesses the advantage of higher abundance compared to ctDNA, thereby improving sensitivity for cancer diagnostics⁴³. Detecting EV mRNA can also be practical for certain targets. For example, *EGFR*^{WT} deletion involves variable breakpoints in the *EGFR* gene⁵⁰, which complicates the design of specific ctDNA assays. Alternatively, EV mRNA analysis can target the conserved *EGFR*^{WT} transcript sequence, facilitating a more straightforward assay. In comparison, ctDNA analyses can provide molecular insights unavailable at the transcript level, such as promoter alterations⁵¹ and methylation patterns⁵², which can enhance diagnostic accuracy and indicate the tumor's tissue of origin. Conveniently, both EV nucleic acids and ctDNA can be extracted from plasma using the same isolation process to expedite these synergistic analyses⁴³.

We envision further technical improvements to enhance SCOPE's preclinical and clinical impact. One straightforward approach is to encompass other major mutations for wider cancer coverage^{53,54}. These targets could include *PIK3CA* (H1047R; E545K), *TP53* (R175H; R273C; G245S), *NRAS* (Q61R; Q61K), *BRAF* (V600E; V600K; V600R) and *EGFR* (L858R; E746_A750del). Additionally, the panel could be tailored for treatment-related targets, such as *MGMT* for temozolomide resistance¹⁴, *EGFR* (T790M) for resistance to EGFR tyrosine kinase inhibitors (TKIs)⁵⁵ and *FGFR3* (R248C; S249C; G370C; Y373C) for sensitivity to FGFR TKI treatments⁵⁶. SCOPE's high selectivity will help design these probes. In addition, the assay's 'one-target-per-tube' format will effectively eliminate potential crosstalk between probes within a reaction vessel; probe optimization can, thus, focus on attaining similar analytical performance among different probes. For example, we prepared

BRAF mutation probes (Supplementary Fig. 24) within 2 weeks, demonstrating the assay's efficiency for rapid probe development.

We can also incorporate protein detection into the assay workflow, capitalizing on the inherent advantage of EVs as a multi-cargo carrier. Detecting proteins will complement mRNA analyses, establishing greater connections between tumor genotypes and phenotypes. Protein markers can offer additional insights into tumors' tissue of origin⁵⁷, crucial information that is often difficult to obtain solely from driver gene mutations. We propose using aptamers or DNA-barcoded antibodies as affinity ligands, which will make protein assays compatible with SCOPE. We can further explore employing orthogonal Cas proteins that exhibit distinct preferences for motif sequences during *trans*-cleavage⁵⁸. By designing signal templates that are fluorescently distinguishable and compatible with each Cas variant, it would be feasible to detect multiple targets (4–6) in a single reaction. In addition, adapting SCOPE to the digital PCR format (that is, sample partitioning) would be an intriguing direction for absolute target quantification and further enhancements in VAF sensitivity. With these advancements, SCOPE would be positioned as a powerful liquid biopsy tool for precision oncology, enabling comprehensive and reliable molecular characterization of tumors in a single, cohesive and accessible platform.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-024-02426-6>.

References

- Ignatiadis, M., Sledge, G. W. & Jeffrey, S. S. Liquid biopsy enters the clinic—implementation issues and future challenges. *Nat. Rev. Clin. Oncol.* **18**, 297–312 (2021).
- Heitzer, E., Haque, I. S., Roberts, C. E. S. & Speicher, M. R. Current and future perspectives of liquid biopsies in genomics-driven oncology. *Nat. Rev. Genet.* **20**, 71–88 (2019).
- Killingsworth, B., Welsh, J. A. & Jones, J. C. EV translational horizons as viewed across the complex landscape of liquid biopsies. *Front. Cell Dev. Biol.* **9**, 556837 (2021).
- Parikh, A. R. et al. Liquid versus tissue biopsy for detecting acquired resistance and tumor heterogeneity in gastrointestinal cancers. *Nat. Med.* **25**, 1415–1421 (2019).
- Pantel, K. & Alix-Panabières, C. Liquid biopsy and minimal residual disease—latest advances and implications for cure. *Nat. Rev. Clin. Oncol.* **16**, 409–424 (2019).
- Siravegna, G., Marsoni, S., Siena, S. & Bardelli, A. Integrating liquid biopsies into the management of cancer. *Nat. Rev. Clin. Oncol.* **14**, 531–548 (2017).
- Shao, H. et al. New technologies for analysis of extracellular vesicles. *Chem. Rev.* **118**, 1917–1950 (2018).
- Im, H. et al. Label-free detection and molecular profiling of exosomes with a nano-plasmonic sensor. *Nat. Biotechnol.* **32**, 490–495 (2014).
- Jo, A. et al. Inaugurating high-throughput profiling of extracellular vesicles for earlier ovarian cancer detection. *Adv. Sci.* **10**, e2301930 (2023).
- Dixon, A. C., Dawson, T. R., Di Vizio, D. & Weaver, A. M. Context-specific regulation of extracellular vesicle biogenesis and cargo selection. *Nat. Rev. Mol. Cell Biol.* **24**, 454–476 (2023).
- Skog, J. et al. Glioblastoma microvesicles transport RNA and proteins that promote tumour growth and provide diagnostic biomarkers. *Nat. Cell Biol.* **10**, 1470–1476 (2008).
- Vitale, S. R. et al. Detection of tumor-derived extracellular vesicles in plasma from patients with solid cancer. *BMC Cancer* **21**, 315 (2021).
- van de Haar, J. et al. Codon-specific *KRAS* mutations predict survival benefit of trifluridine/tipiracil in metastatic colorectal cancer. *Nat. Med.* **29**, 605–614 (2023).
- Shao, H. et al. Chip-based analysis of exosomal mRNA mediating drug resistance in glioblastoma. *Nat. Commun.* **6**, 6999 (2015).
- Park, J. et al. An integrated magneto-electrochemical device for the rapid profiling of tumour extracellular vesicles from blood plasma. *Nat. Biomed. Eng.* **5**, 678–689 (2021).
- Noerholm, M. et al. RNA expression patterns in serum microvesicles from patients with glioblastoma multiforme and controls. *BMC Cancer* **12**, 22 (2012).
- Wei, Z. et al. Coding and noncoding landscape of extracellular RNA released by human glioma stem cells. *Nat. Commun.* **8**, 1145 (2017).
- Hong, J. S., Son, T., Castro, C. M. & Im, H. CRISPR/Cas13a-based microRNA detection in tumor-derived extracellular vesicles. *Adv. Sci.* **10**, e2301766 (2023).
- Yan, H. et al. A one-pot isothermal Cas12-based assay for the sensitive detection of microRNAs. *Nat. Biomed. Eng.* **7**, 1583–1601 (2023).
- Ramshani, Z. et al. Extracellular vesicle microRNA quantification from plasma using an integrated microfluidic device. *Commun. Biol.* **2**, 189 (2019).
- Li, Y. et al. EV-origin: enumerating the tissue-cellular origin of circulating extracellular vesicles using exLR profile. *Comput. Struct. Biotechnol. J.* **18**, 2851–2859 (2020).
- Pickar-Oliver, A. & Gersbach, C. A. The next generation of CRISPR–Cas technologies and applications. *Nat. Rev. Mol. Cell Biol.* **20**, 490–507 (2019).
- Kaminski, M. M., Abudayyeh, O. O., Gootenberg, J. S., Zhang, F. & Collins, J. J. CRISPR-based diagnostics. *Nat. Biomed. Eng.* **5**, 643–656 (2021).
- Abudayyeh, O. O. & Gootenberg, J. S. CRISPR diagnostics. *Science* **372**, 914–915 (2021).
- Potapov, V. & Ong, J. L. Examining sources of error in PCR by single-molecule sequencing. *PLoS ONE* **12**, e0169774 (2017).
- Kebschull, J. M. & Zador, A. M. Sources of PCR-induced distortions in high-throughput sequencing data sets. *Nucleic Acids Res.* **43**, e143 (2015).
- Gootenberg, J. S. et al. Nucleic acid detection with CRISPR–Cas13a/C2c2. *Science* **356**, 438–442 (2017).
- Truong, N. P., Jia, Z., Burges, M., McMillan, N. A. & Monteiro, M. J. Self-catalyzed degradation of linear cationic poly(2-dimethylaminoethyl acrylate) in water. *Biomacromolecules* **12**, 1876–1882 (2011).
- Maslak, M. & Martin, C. T. Kinetic analysis of T7 RNA polymerase transcription initiation from promoters containing single-stranded regions. *Biochemistry* **32**, 4281–4285 (1993).
- Avaro, A. S. & Santiago, J. G. Uncertainty quantification of Michaelis-Menten kinetic rates and its application to the analysis of CRISPR-based diagnostics. *Angew. Chem. Int. Ed. Engl.* **61**, e202209527 (2022).
- Newman, A. M. et al. An ultrasensitive method for quantitating circulating tumor DNA with broad patient coverage. *Nat. Med.* **20**, 548–554 (2014).
- Diehl, F. et al. Detection and quantification of mutations in the plasma of patients with colorectal tumors. *Proc. Natl Acad. Sci. USA* **102**, 16368–16373 (2005).
- Li, R. et al. Therapeutically reprogrammed nutrient signalling enhances nanoparticulate albumin bound drug uptake and efficacy in *KRAS*-mutant cancer. *Nat. Nanotechnol.* **16**, 830–839 (2021).
- Pfirschke, C. et al. Immunogenic chemotherapy sensitizes tumors to checkpoint blockade therapy. *Immunity* **44**, 343–354 (2016).

35. DuPage, M., Dooley, A. L. & Jacks, T. Conditional mouse lung cancer models using adenoviral or lentiviral delivery of Cre recombinase. *Nat. Protoc.* **4**, 1064–1072 (2009).
36. Mendiratta, G. et al. Cancer gene mutation frequencies for the U.S. population. *Nat. Commun.* **12**, 5961 (2021).
37. Welsh, J. A. et al. Minimal information for studies of extracellular vesicles (MISEV2023): from basic to advanced approaches. *J. Extracell. Vesicles* **13**, e12404 (2024).
38. Weller, M. et al. EANO guidelines on the diagnosis and treatment of diffuse gliomas of adulthood. *Nat. Rev. Clin. Oncol.* **18**, 170–186 (2021).
39. Guo, G. et al. Ligand-independent EGFR signaling. *Cancer Res.* **75**, 3436–3441 (2015).
40. Chen, W. W. et al. BEAMing and droplet digital PCR analysis of mutant IDH1 mRNA in glioma patient serum and cerebrospinal fluid extracellular vesicles. *Mol. Ther. Nucleic Acids* **2**, e109 (2013).
41. SongTao, Q. et al. IDH mutations predict longer survival and response to temozolomide in secondary glioblastoma. *Cancer Sci.* **103**, 269–273 (2012).
42. Batool, S. M. et al. Highly sensitive *EGFRvIII* detection in circulating extracellular vesicle RNA of glioma patients. *Clin. Cancer Res.* **28**, 4070–4082 (2022).
43. Krug, A. K. et al. Improved EGFR mutation detection using combined exosomal RNA and circulating tumor DNA in NSCLC patient plasma. *Ann. Oncol.* **29**, 700–706 (2018).
44. Srivastava, P. & Prasad, D. Isothermal nucleic acid amplification and its uses in modern diagnostic technologies. *3 Biotech* **13**, 200 (2023).
45. Ueda, T., Tohda, H., Chikazumi, N., Eckstein, F. & Watanabe, K. Phosphorothioate-containing RNAs show mRNA activity in the prokaryotic translation systems in vitro. *Nucleic Acids Res.* **19**, 547–552 (1991).
46. Lane, R. E., Korbie, D., Trau, M. & Hill, M. M. Optimizing size exclusion chromatography for extracellular vesicle enrichment and proteomic analysis from clinically relevant samples. *Proteomics* **19**, e1800156 (2019).
47. Lee, K. et al. Multiplexed profiling of single extracellular vesicles. *ACS Nano* **12**, 494–503 (2018).
48. Zhang, J. et al. Engineering a tunable micropattern-array assay to sort single extracellular vesicles and particles to detect RNA and protein in situ. *J. Extracell. Vesicles* **12**, e12369 (2023).
49. Schraa, S. J. et al. Circulating tumor DNA guided adjuvant chemotherapy in stage II colon cancer (MEDOCC-CrEATE): study protocol for a trial within a cohort study. *BMC Cancer* **20**, 790 (2020).
50. Koga, T. et al. Mapping of genomic *EGFRvIII* deletions in glioblastoma: insight into rearrangement mechanisms and biomarker development. *Neuro Oncol.* **20**, 1310–1320 (2018).
51. Muralidharan, K. et al. TERT promoter mutation analysis for blood-based diagnosis and monitoring of gliomas. *Clin. Cancer Res.* **27**, 169–178 (2021).
52. Shen, S. Y. et al. Sensitive tumour detection and classification using plasma cell-free DNA methylomes. *Nature* **563**, 579–583 (2018).
53. Chang, M. T. et al. Identifying recurrent mutations in cancer reveals widespread lineage diversity and mutational specificity. *Nat. Biotechnol.* **34**, 155–163 (2016).
54. Miller, M. L. et al. Pan-cancer analysis of mutation hotspots in protein domains. *Cell Syst.* **1**, 197–209 (2015).
55. Sequist, L. V. et al. Genotypic and histological evolution of lung cancers acquiring resistance to EGFR inhibitors. *Sci. Transl. Med.* **3**, 75ra26 (2011).
56. Krook, M. A. et al. Fibroblast growth factor receptors in cancer: genetic alterations, diagnostics, therapeutic targets and mechanisms of resistance. *Br. J. Cancer* **124**, 880–892 (2021).
57. Cohen, J. D. et al. Detection and localization of surgically resectable cancers with a multi-analyte blood test. *Science* **359**, 926–930 (2018).
58. Gootenberg, J. S. et al. Multiplexed and portable nucleic acid detection platform with Cas13, Cas12a, and Csm6. *Science* **360**, 439–444 (2018).

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Methods

Materials

All oligonucleotides used in this study were purchased from Integrated DNA Technologies. The specific sequences of the oligonucleotides are listed in Supplementary Table 1 (SCOPE) and Supplementary Table 5 (RT-PCR). Monarch RNA Lysis Buffer, Monarch RNase A, RNase inhibitor (Murne), Proteinase K, DNase I, HiScribe T7 High Yield T7 RNA synthesis kit, Luna Universal One-Step RT-qPCR kit and Quick-Load Purple Low Molecular Weight DNA Ladder were obtained from New England Biolabs. Qubit RNA HS Assay kit and DEPC-DW were purchased from Thermo Fisher Scientific. The RNeasy Micro kit was purchased from Qiagen, and the Total Exosome RNA & Protein Isolation kit was from Invitrogen. GelRed nucleic acid stain (41003) was sourced from Biotium. All other chemicals used in the experiments were of analytical grade and used without further purification.

Expression and purification of LwaCas13a

We prepared the LwaCas13a protein through bacterial expression and subsequent purification. A plasmid DNA carrying the LwaCas13a encoding sequence (Addgene, plasmid 90097) was introduced into the *Escherichia coli* strain Rosetta 2 (DE3) pLysS (Millipore Sigma). The transformation process relied on the well-established heat shock transformation method, specifically optimized for bacterial expression. After transformation, protein expression was induced using isopropylthio- β -galactoside (0.5 mM; Millipore Sigma). LwaCas13a proteins were then isolated through a systematic multi-step process, commencing with cell lysis and initial nickel-nitrilotriacetic acid (Ni-NTA; Thermo Fisher Scientific) purification. The solution underwent the first buffer exchange to a small ubiquitin-like modifier (SUMO) cleavage buffer (30 mM Tris-HCl at pH 8.0, 500 mM NaCl and 1 mM dithiothreitol (DTT)). This was followed by treatment with SUMO protease (Thermo Fisher Scientific), a second round of Ni-NTA purification and a final buffer exchange to the storage buffer (50 mM Tris-HCl at pH 7.5, 600 mM NaCl and 2 mM DTT). The purified LwaCas13a proteins were stored with 5% glycerol (Millipore Sigma) and a protease inhibitor (Roche) at -80°C until use. To verify the successful purification of LwaCas13a proteins, the products obtained from each purification step were subjected to SDS-PAGE followed by Ag staining (Supplementary Fig. 24).

Preparation of pDMAEA-coated PCR tubes

We used PCR tubes (diameter, 6 mm) purchased from Bio-Rad. Tert-butyl peroxide (TBPO, 98%) and dimethylaminoethyl acrylate (DMAEA, 98%) were purchased from Merck. The DMAEA monomer and TBPO were vaporized through heat exposure at 35°C and 25°C , respectively. Subsequently, the vaporized substances were introduced into an initiated chemical vapor deposition (iCVD) chamber (160 mTorr). The filament temperature of the iCVD was set at 140°C to generate TBPO radicals. The tubes were held at 30°C to facilitate vapor absorption. The pDMAEA thin film was produced as a result of the vapor-phase free radical polymerization, facilitated by a filament temperature of 140°C .

Characterization of pDMAEA hydrolysis

For NMR analysis, we dissolved pDMAEA in CDCl_3 or phosphate D_2O buffer at pH 8.0. Subsequently, 1H NMR spectra were collected using a Bruker Avance Neo 600-MHz spectrometer (Bruker BioSpin). The degree of hydrolysis was determined by comparing the integrated signals for the methylene of the ester in pDMAEA (-4.41 ppm) and the alcohol of dimethylaminoethanol (DMAE) (-4.00 ppm). For Fourier transform infrared (FTIR) analysis, we collected the spectra in the wavenumber range of $400\text{--}4,000\text{ cm}^{-1}$ and at a resolution of 1.4 cm^{-1} , using an Alpha instrument from Bruker Optics. The zeta-potential values were determined using a Nano Zetasizer 3600 (Malvern) equipped with a He-Ne laser source operating at 633 nm .

Characterization of pDMAEA tube

To evaluate the RNA extraction efficiency of the pDMAEA-coated tube, we added $100\text{ }\mu\text{l}$ of initial RNA (3, 30, 60 or 100 ng) solution to the tube and let the solution react for 10 min. We then centrifuged the tube for 5 min ($123g$), collected the supernatant and quantified the RNA concentration (supernatant) using the Qubit RNA HS Assay kit. For the comparison, we processed the same RNA samples ($100\text{ }\mu\text{l}$ with initial RNA amounts of 3, 30, 60 and 100 ng) using the RNeasy Micro Qiagen kit based on silica columns. Following the RNeasy protocol, the concentration of the extracted RNA was measured using the Qubit RNA HS Assay kit.

Construction of SCOPE device

We designed the device using design software (Autodesk, Inventor). The device's primary components were fabricated via computer numerical control (CNC) machining on Al alloy (AL6061) and injection molding with polycarbonate.

Heating block. The block was machined on an Al block to hold 16 standard PCR tubes (diameter, 6 mm). We used two polyimide heaters to heat the block: a high-powered 144-W main heater and a 24-W top heater. These heaters were driven by field-effect transistors connected to a 24-V power supply. The temperature of each heater was continually monitored using a $10\text{-k}\Omega$ thermistor (TDK, NTCG103JF103F) and was maintained within a 0.3°C tolerance through feedback control.

Fluorescent optics. The device had an optical module consisting of two independent units, each configurable for a different fluorescent channel. For the green fluorescent channel (excitation 495 nm ; emission 520 nm) used in this study, the components in the unit were a blue light-emitting diode (Cree LED, C503B-BAN-CZ0A0451), a collimating lens (Roithner Lasertechnik, LC12), an excitation band-pass filter (Semrock, FF01-474/27), a dichroic mirror (Semrock, Di02-R488), an emission band-pass filter (Semrock, FF01-515/30) and two focusing lenses (Roithner Lasertechnik, LC7 and GS7020-2). The fluorescent emission from the sample was detected by a photodiode that had a broad spectral response range from 320 nm to $1,100\text{ nm}$ (Hamamatsu, S2386-44K). The current from the photodiode was converted into a voltage signal by a transimpedance amplifier (Texas Instruments, OPA320) and further boosted by a non-inverting amplifier (Analog Devices, AD8605). The resultant analog signal was converted into a digital format by a 16-bit analog-to-digital converter (Texas Instruments, ADS1115) that collected data at a 62.5 samples per second sampling rate.

Linear actuator. We mounted the optical module on a linear actuator to scan the module across 16 samples for fluorescent detection. The actuator was powered by a stepping motor (Kiatronics, 28BYJ-48). The position of the optical module was initialized by moving the module until it contacted a limit switch.

Microcontroller unit. We programmed an ESP32 microcontroller unit (MCU) (Espressif) to operate the device. The MCU processed various data streams, including fluorescent measurement results, scanning locations and heater temperatures. It also communicated with an external tablet through a Bluetooth connection. The MCU firmware was written in C++ (gcc version 12.1).

SCOPE app

The software was an Android app with an intuitive graphical user interface. We wrote the app using JAVA and the Android software development kit. The app saved all fluorescent measurements and test settings in a standardized CSV format to ensure compatibility across various operating systems.

Cell culture

We purchased a panel of cell lines from the American Type Culture Collection (ATCC) and grew them in the vendor-recommended media: H2228 (CellGro, RPMI 1640); A549 (ATCC, F-12K); CCD-18 (Corning, DMEM); Colo205 (RPMI 1640); HCT116 (CellGro, McCoy's 5A); HT29 (CellGro, McCoy's 5A modified with 2% NaHCO₃); LS174T and LS123 (CellGro, Eagle's Minimum Essential Medium); and SW480 and SW620 (Corning, DMEM). The KP1.9 cell line was derived from tumors of the KP mouse model on a C57BL/6 congenic background and was a generous gift from A. Zippelius (University Hospital Basel, Switzerland). KP1.9 cells were cultured in Iscove's DMEM medium (Corning). For GBM cell lines, we purchased GLI36 cells from the ATCC and generated GLI36VIII and GLI36-R132H cell lines through lentivirus transduction. Both GLI36 and its variants were cultured in DMEM (Corning). All media were supplemented with 10% FBS and penicillin–streptomycin (Corning). All cell lines were tested and determined to be free of mycoplasma contamination (Lonza; MycoAlert Mycoplasma Detection Kit, LT07-418).

EV preparation via SEC

We cultured cells at passages 1–15 in a vesicle-depleted medium (with 5% depleted FBS) for 48 h. We then collected culture media from approximately 10⁷ cells and removed cellular debris through two successive rounds of centrifugation (10,000g, 3 min). To isolate EVs, we first concentrated the conditioned culture medium by processing it with a centrifugal filter (Millipore Sigma, Centricon Plus-70). Subsequently, we loaded the concentrated medium (0.5 ml) to a size exclusion column (Izon, SP1) and collected EV fractions (F7–F9; 1.5 ml) in low-protein-binding microcentrifuge tubes (Thermo Fisher Scientific, 90410). Collected EVs were resuspended in 1× PBS buffer and stored at –80 °C. We estimated EV concentrations in these stock solutions via nanoparticle tracking analysis (NTA). For each sample, we acquired three 30-s video clips (Malvern, NanoSight LM10) and analyzed them using NTA software (version 3.2).

EV preparation via ultracentrifugation

Cells were cultured until they reached approximately 70% confluency. At this stage, the supernatant was removed and replaced with F-12K Medium (ATCC, 30-2004) containing EV-depleted FBS. After a 24-h cell culture, the culture media supernatant was processed through sequential centrifugation. Initially, the supernatant underwent centrifugation at 300g for 10 min and then at 2,000g for 10 min using a centrifuge to eliminate cells and debris. Subsequently, the supernatant was subjected to ultracentrifugation (Beckman, Optima L-100K) at 4 °C and 10,000g for 30 min to remove proteins. The supernatant was ultracentrifuged at 4 °C and 100,000g for 70 min to pellet EVs. The pellets were resuspended in cold PBS and ultracentrifuged again at 100,000g for 70 min at 4 °C to eliminate residual impurities. Finally, the EV pellets were collected, resuspended in PBS (1.5 ml) and stored at –80 °C. For SCOPE analysis, we used 100 µl of this sample.

Western blotting

EVs were enriched from a plasma sample (3 ml) obtained from a patient with CRC. The plasma was initially centrifuged at 2,000g for 3 min to remove cellular debris, and the resulting supernatant was processed using SEC (Izon, qEV). The isolated EVs were concentrated to 100 µl using Amicon Ultra-2 10K filters (Millipore Sigma). For western blot analysis, the concentrated EVs were lysed with ice-cold 10× RIPA buffer (Abcam) for 30 min, followed by centrifugation at 14,000g for 10 min at 4 °C. Protein concentration was determined via a BCA assay, and the lysates were analyzed via 10% SDS-PAGE. The following primary antibodies were used: Alix (Invitrogen, JM85-31; 1:1,000), biotinylated CD63 (AHN16.1/46-4-5, Ancell; 1:1,000) and Histone H2B (D2H6, Cell Signaling Technology; 1:1,000). Before use, the Alix and Histone H2B antibodies were biotinylated using sulfo-NHS-Biotin (Thermo Fisher Scientific) in accordance with the manufacturer's protocol.

Secondary detection was subsequently conducted using HRP-conjugated streptavidin (R&D Systems; 1:2,000).

SCOPE assay

We placed an EV sample (100 µl) and a lysis buffer (10 µl) in a pDMAEA-coated tube. The lysis buffer consisted of proteinase K (0.8 U per reaction), RNase inhibitor (1 U per reaction) and DNase I (2 U per reaction) in Tris-EDTA (TE; pH 8.0). After incubating the mixture at 25 °C for 10 min, we centrifuged the sample (123g, 5 min) and discarded the supernatant, leaving about 1 µl of the lysate in the tube. We next added the SCOPE master mix (20 µl) to the same tube and let the mixture react (40 °C, 30 min) inside the SCOPE device. The master mix had the following components: 5 µM LwaCas13a, 10 µM crRNA, 50 µM Signal template, 16 U µl^{–1} RNase inhibitor, 10 mM deoxyribonucleotide triphosphate mixture (ATP, GTP, UTP and CTP), 0.1 µM RNA polymerase, 0.5× Cas13a cleavage buffer (10 mM HEPES-Na at pH 6.8, 25 mM KCl, 2.5 mM MgCl₂ and 2.5% glycerol) and 0.5× RNAPol buffer (20 mM Tris-HCl at pH 7.9, 3 mM MgCl₂, 0.5 mM DTT and 1 mM spermidine). Throughout the SCOPE reaction, we monitored the fluorescent signals using the device.

Quantitative PCR

We used RT-qPCR for comparison with SCOPE. We used either synthetic RNA samples (1 µl) or EV RNA samples (1 µl) as prepared in the SCOPE assay. A sample was mixed with the master mix (20 µl) containing 1× Luna Universal One-Step Reaction Mix, 1× Luna WarmStart RT Enzyme Mix and 400 nM primers. We then carried out reactions on a CFX Connect 96 Real-Time PCR System (Bio-Rad) using the following thermal cycling schedule: 55 °C, 10 min (reverse transcription) and 95 °C, 1 min (RTase inactivation), followed by 40 cycles of 95 °C (10 s), 60 °C (30 s) and fluorescent measurements.

Gel electrophoresis

We resolved 20 µl of the reaction mixture on a gel composed of 15% polyacrylamide. We used a 1× TE solution as a running buffer and applied a constant voltage of 120 V throughout the electrophoresis process (2 h). Upon the completion of this phase, the polyacrylamide gels were subjected to staining with GelRed. Subsequently, we scanned the stained gels using a GelDoc Go imaging system (Bio-Rad).

Comparison of SEC-processed and direct plasma samples for the SCOPE assay

A plasma sample from a patient with CRC was divided into two aliquots of 40 µl each. One aliquot underwent SEC to yield a final volume of 600 µl of EV isolate, from which RNA was extracted using the Total Exosome RNA & Protein Isolation Kit (Invitrogen, 4478545). The remaining plasma aliquot was processed for RNA extraction using QIAamp RNA Blood Mini Kit (Qiagen, 52304) in accordance with the manufacturer's instructions. The extracted RNA was then quantified, and the final concentration was adjusted to 1 ng µl^{–1}. The extracted RNA samples (50 µl) were then used for the SCOPE assay.

Comparative experiments with RNase treatment

We used 40 µl of plasma samples from patients with CRC and isolated EVs via SEC. The final sample volume was 600 µl after SEC operation. We mixed 100 µl of this EV isolate with 3 µl of RNase A (New England Biolabs, T3018L). The mixture was incubated at 25 °C for 5 min. Subsequently, the mixture was filtered through a 30-kDa Amicon Ultra centrifugal filter (Millipore, UFC503008) to remove RNase A. A control sample underwent the same processes, with RNase A replaced by PBS. All prepared samples were then lysed and subjected to the SCOPE assay.

EV profiling from mouse samples

We conducted the animal study following the guidelines provided by the Institutional Animal Care and Use Committee of Massachusetts

General Hospital (protocol 2019N000133). We used 129S1 mice, which carried the Cre-activatable conditional *Kras*^{LSL-G12D/+}; *Trp53*^{fllox/fllox} (KP mice) as an autochthonous model of lung adenocarcinoma^{33,35}. Mice were fed with autoclaved food and water and maintained in ventilated cages in a light/dark cycle, temperature (18–23 °C) and humidity (40–60%) controlled pathogen-free vivarium at Massachusetts General Hospital. To initiate the tumor growth, we intranasally instilled adenoviruses (3×10^7 particles) expressing Cre recombinase (Ad-Cre, University of Iowa Viral Vector Core). We collected blood samples (~50 µl) from the mice before the virus instillation and weekly thereafter, using the retro-orbital bleeding method. Mice were monitored for signs of disease progression and distress, including according to their body condition score and body weight. To collect plasma, we centrifuged blood samples (2,000g, 15 min) at 4 °C and recovered supernatant. Plasma samples were stored at –80 °C until use. For EV isolation, we diluted plasma samples (40 µl) in PBS to 100 µl and processed them using size exclusion columns (Izon, SP2) as described above.

EV profiling with clinical samples

The study was approved by the institutional review board (IRB) of Massachusetts General Hospital (protocols 2017P001581, 2019P003441 and 2019P003472) and the IRB of Kyungpook National University Chilgok Hospital (protocol KNUCH 2021-10-025). The procedures followed were per the respective institutional guidelines. Informed consent was obtained from all participants. We drew peripheral blood samples (~10 ml) from patients into blood collection tubes (BD, Vacutainer 367525). Samples were then centrifuged (2,000g, 3 min) to separate plasma from red blood cells and buffy coat and stored at –80 °C until use. For EV isolation, stored plasma samples (40 µl) were diluted in PBS (60 µl) and processed with size exclusion columns (Izon, SP1).

Tissue analysis

Patients with CRC. Pathologic evaluation and *KRAS* analysis of CRC tissues were performed by the Department of Pathology at the Kyungpook National University Chilgok Hospital. Formalin-fixed, paraffin-embedded (FFPE) tissue blocks were used for DNA extraction. PNAclamp *KRAS* Mutation Detection kits, version 4 (Panagene), capable of detecting 40 *KRAS* mutation variants across codons 12, 13, 59, 61, 117 and 146, were employed for assays following the manufacturer's instructions. The Massachusetts General Hospital Pathology Service analyzed pathologic tissues from patients with glioma. The analysis involved the use of various methods, including IHC for EGFR, Snapshot NGS for *IDH1*, Genexus NGS for *IDH1* and the Solid Fusion assay for *EGFR* amplification.

IHC

Lung tissue from KP mice. We anesthetized mice with isoflurane and perfused them with 4% paraformaldehyde (PFA) through the trachea and by the tail vein. Lungs were further fixed for 24 h, washed with PBS and soaked in 30% sucrose in PBS for 24 h. The lung tissue was then embedded in optimum cutting temperature (OCT) compound (Sakura Finetek) and snap frozen in an isopentane bath on dry ice. Once the tissue was frozen, we prepared serial 10-µm-thick sections using a cryostat and placed them on glass slides. Samples were then sent to the Histopathology Research Core at Massachusetts General Hospital for staining: hematoxylin and eosin (H&E) and RAS G12D (GeneTex; GTX635362, 25× dilution).

Tissue from patients with CRC. IHC was performed on representative FFPE blocks using an automatic IHC staining instrument (Ventana Medical Systems, BenchMark XT) according to the manufacturer's protocols. In brief, 4-µm FFPE sections were transferred to poly-L-lysine-coated slides and dried in an oven at 65 °C for 2 h. After deparaffinization

in xylene and rehydration through a series of graded alcohols, we retrieved antigens by boiling samples via microwave in 10 mM sodium citrate buffer (pH 6.0). Sections were then incubated with a primary antibody against RAS G12D (GeneTex; GTX635362, 200× dilution) overnight at 4 °C. Slides were then visualized using the UltraView Universal DAB kit (Ventana Medical Systems) and counterstained with Harris hematoxylin for nucleus. RAS G12D protein was considered positive when IHC staining was positive in the cytoplasm of tumor cells and was not expressed in normal non-neoplastic colonic epithelium cells.

Statistical analysis

We used GraphPad Prism version 10.2.3 (GraphPad Software) or R version 4.2.2 for statistical analyses. For two-group comparisons, we used an unpaired, two-sided *t*-test. For all statistical tests, *P* values less than 0.05 were considered significant. Details on data presentation and the sample size are included in the figure legends.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The mRNA target sequences were obtained from the National Center for Biotechnology Information's Reference Sequence database (accession codes: [NC_000002.12](#), [NC_00007.14](#) and [NC_000012.12](#)). The primary data supporting assay characterization are accessible in a source data file and in Supplementary Information. Raw patient datasets generated and analyzed during this study are available from the corresponding authors upon reasonable request, subject to approval from the institutional review board of Massachusetts General Hospital (MGH) and the MGH Innovation Office. Data access requests will be considered from academic investigators without relevant conflicts of interest, for non-commercial use, who agree to non-distribution terms. Source data are provided with this paper.

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Author contributions

J.S., T.K., C.M.C. and H.L. developed the concept and designed the study. J.S., M.H.C., H.C., J.-S.H. and J.J. performed the experiments. M.H.C., H.L. and M.A.M. designed and analyzed the animal model studies. Y.S. and S.G.I. prepared the pDMAEA-coated tube. S.W.L., H.C.N., T.H.Y., J.C.S. and H.L. designed the SCOPE device. E.E., D.G.Y., B.C.B. and L.B. collected blood samples from patients with GBM and analyzed their tissue samples. Y.K., G.-S.C., J.S.P., A.N.S. and S.Y.P. collected blood samples from patients with CRC and analyzed their tissue samples. J.S., T.K., C.M.C. and H.L. wrote the paper, with input from all authors.

Competing interests

J.S., T.K., C.M.C. and H.L. declare the filing of a provisional patent that was assigned to and handled by Massachusetts General Hospital and the Korea Research Institute of Bioscience and Biotechnology. M.A.M. declares research support from Ionis Pharmaceuticals, Genentech and Pfizer, all of which are unrelated to the present paper. The other authors declare no competing interests.

Additional information

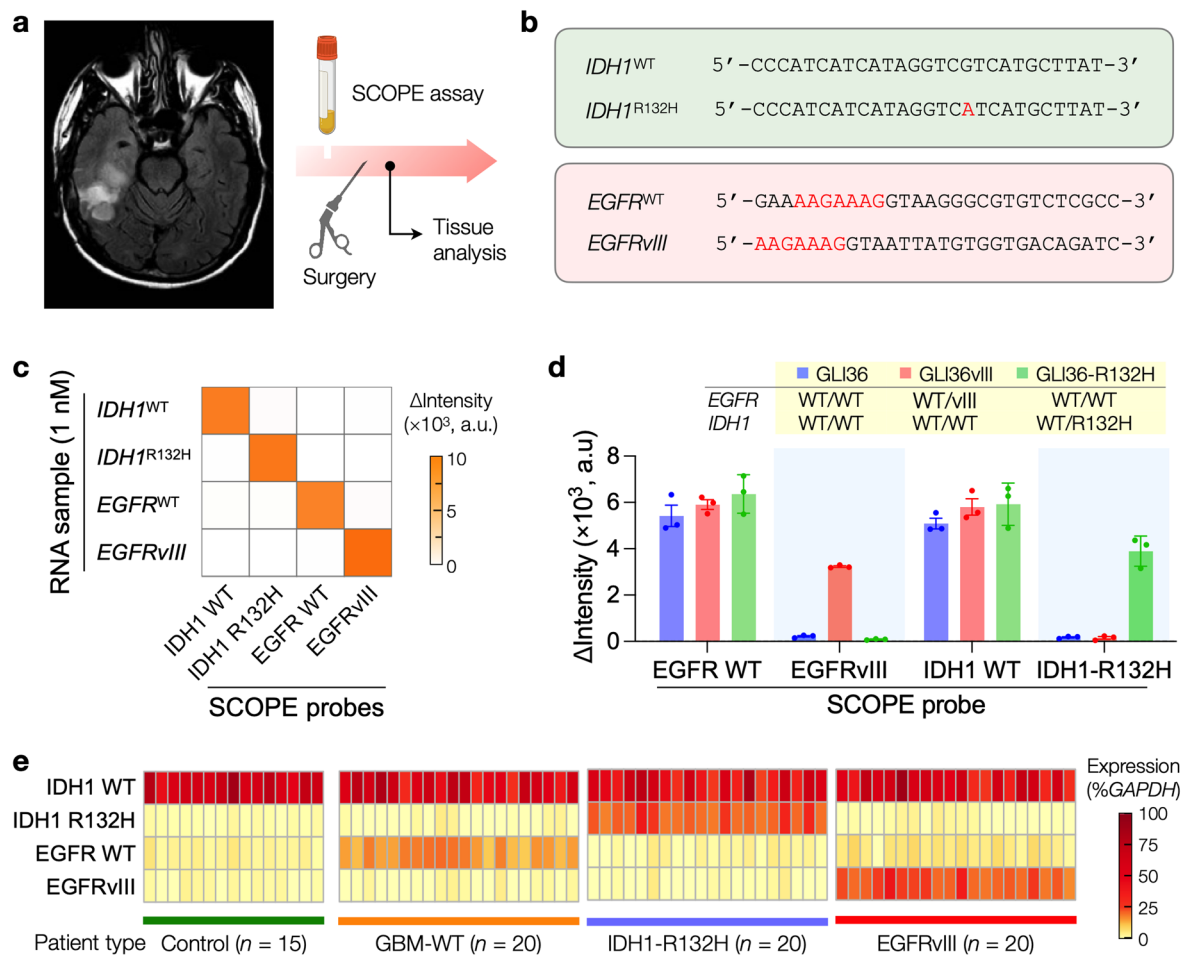
Extended data is available for this paper at <https://doi.org/10.1038/s41587-024-02426-6>.

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Extended Data Fig. 1 | Stratifying glioma patients. (a) SCOPE analyzed plasma samples collected from radiologically confirmed glioma patients. Tissue samples were used for clinical pathology. (b) Target RNA sequences for glioma-SCOPE analyses. *IDH1* probes were designed to detect wild-type (WT) and single nucleotide mutation (R132H), and *EGFR* probes to detect WT as well as the variant III (*EGFR*^{vIII}) resulting from genomic deletion of exons 2–7 in the *EGFR* gene. (c) The specificity of glioma-SCOPE probes was evaluated. The designed probes achieved a high signal contrast (>30) between on-target and off-target samples. Synthetic RNA samples (1 nM) were used. The heatmap displays mean values from technical triplicate measurements. a.u., arbitrary unit. (d) Application of

SCOPE to profile EVs for *IDH1* (WT, R132H) and *EGFR* (WT, vIII). EVs were harvested from glioma cell lines. The results confirmed that EVs reflected the genotype of parent glioma cells. Data are shown as mean $\pm$ s.d. from biological triplicates. a.u., arbitrary unit. (e) Plasma EVs were analyzed via SCOPE for *IDH1* (WT, R132H) and *EGFR* (WT, vIII). Control samples were from healthy donors ($n = 15$). Glioma samples were from patients with different genotypes: *EGFR* amplification (GBM-WT; $n = 20$), *IDH1*-R132H mutation ($n = 20$), and *EGFR*vIII mutation ($n = 20$). The SCOPE assay identified glioma patients and correctly stratified them according to their molecular genotypes. The heatmap shows mean values from technical triplicate measurements.

Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No software was used for data collection.
Data analysis	Programming language: R 4.2.2 R Core Team https://www.r-project.org/ and C++ (gcc version 12.1) https://gcc.gnu.org/ Statistical analysis and graphing of data: Prism v10.2.3 GraphPad https://www.graphpad.com/scientificsoftware/prism/ Nanoparticle Tracking Analysis: NTA software v3.2 https://www.malvernpanalytical.com/en/support/product-support/nanosight-range

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- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The mRNA target sequences were obtained from the National Center for Biotechnology Information (NCBI) Reference Sequence database (accession codes:

NC_000002.12, NC_00007.14, NC_000012.12). The primary data supporting assay characterization are accessible in a Source Data file and Supplementary Information. Raw patient datasets generated and analyzed during this study are available from the corresponding authors upon request, subject to approval from the Institutional Review Board of Massachusetts General Hospital (MGH) and MGH Innovation Office. Data access requests will be considered from academic investigators without relevant conflicts of interest, for noncommercial use, who agree to non-distribution terms.

Research involving human participants, their data, or biological material

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Reporting on sex and gender

The gender/sex of participants was determined based on self-reported information provided on the study consent form. The study design included both males and females, reflecting the prevalence of colorectal cancer and glioblastoma multiforme (GBM) across sexes.

Reporting on race, ethnicity, or other socially relevant groupings

Race/ethnicity was not considered as an explanatory variable in data analyses. The race/ethnicity of participants was determined based on self-reported information provided on the study consent form.

Population characteristics

Subjects considered for enrollment in this study were evaluated solely on the basis of protocol eligibility criteria. Cancer patients must have a diagnosis of cancer, and the healthy donors are free of cancer. The summary of patient characteristics is given in Supplementary Table 2 and Supplementary Table 3.

Recruitment

We enrolled the first subjects who met the eligibility criteria. Children defined as aged less than 18 were excluded from the study. Otherwise, there were no specific exclusion criteria. Patient recruitment was conducted by treating physicians in accordance with the established eligibility criteria, mitigating the risk of self-selection bias. Informed consent was obtained from all participants, as documented in the manuscript.

Ethics oversight

The study was approved by the Institutional Review Board (IRB) of Massachusetts General Hospital (Protocol 2017P001581, 2019P003441, 2019P003472) and the IRB of Kyungpook National University Chilgok Hospital (Protocol KNUCH 2021-10-025).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

The sample size was not predetermined, but our sample number is powered to determine whether the EV test offers high diagnostic accuracy.

Lung cancer animal study:

In our EV marker measurements with the mouse model, the mean KRASG12D values were 248 ± 21 (s.d.) before tumor induction (week 0) and 909 ± 494 (s.d.) a week after. With a one-sided test and a significance level (α) of 0.05, a sample size of 10 mice would be sufficient to achieve 99% power to detect this effect size (1.89).

CRC and GBM studies:

We used the area under the ROC curve (AUC) as an index of accuracy. Our null hypothesis was that the EV test has an AUC of 0.6 (a poor test), whereas the alternative hypothesis was that the EV test performs significantly better (AUC = 0.8, an accurate test). 1. For the colorectal cancer (CRC) study, the sample size of our cohort (15 non-cancer controls and 107 CRC cases) achieves at least 90% power to detect this AUC difference of 0.2 at a significance level of 5% (one-sided test). 2. For the GBM study, the sample size of our cohort (15 non-cancer controls and 60 GBM cases) achieves at least 87% power to detect this AUC difference of 0.2 at a significance level of 5% (one-sided test).

Data exclusions

No data were excluded from the analyses.

Replication

All samples were measured at least 3 times (technical replica) for each protein marker, and the mean values were used for analyses. All attempts at replication were successful.

Randomization

Subjects meeting the eligibility criteria, which included a confirmed cancer diagnosis, were enrolled on a rolling basis. The allocation process was non-randomized; instead, recruiting physicians stratified patients based on tumor type (colorectal cancer or glioblastoma multiforme) and tissue mutation status (e.g., KRAS wild type or mutant, EGFRV8, IDH1 wild type or mutant). This stratification approach served to control for key covariates, creating homogeneous subgroups and minimizing potential confounding effects.

Blinding

We analyzed the clinical samples using the SCOPE assay while blinded to the clinical diagnostic results (gold standard). The investigators were also blinded to group allocation during data collection and analysis.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Anti RAS G12D (GTX635362, GeneTex); Anti Alix (JM85-31, Invitrogen); Anti CD63 (AHN16.1/46-4-5, Ancell); Anti Histone H2B (D2H6, Cell Signaling Technology).
Validation	All antibodies were commercially obtained, and validation was based on the datasheets from the manufacturer as well as the usage in the field. === Company statements === Anti RAS G12D (GTX635362): The RAS (G12D Mutant) antibody [HL10] is the first commercially available recombinant antibody that demonstrates exceptional specificity by paraffin-embedded immunohistochemistry and western blot for the RAS G12D mutant protein. Anti Alix (JM85-31): This Antibody was verified by Knockdown to ensure that the antibody binds to the antigen stated. Anti CD63 (AHN16.1/46-4-5): Antibody AHN16.1 recognizes the human CD63 molecule of about 50 kd. It also binds to equine (horse) CD63. This mAb specifically promotes adhesion of human melanoma cells in a Ca²⁺ dependant manner. Anti Histone H2B (D2H6): Histone H2B (D2H6) Rabbit mAb recognizes endogenous levels of total H2B protein. This antibody does not cross-react with other histone proteins.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	H2228, A549, CCD18Co, Colo205, HT29, LS123, LS174T, SW480, SW620, HCT116, and GLI36 were from ATCC. GLI36vIII and GLI36-R132H were provided by Dr. Balaj (MGH). KP1.9 mouse lung cancer cells were obtained from Dr. Annette Zippelius (University Hospital Basel).
Authentication	All cell lines were authenticated by the providers. We used cells without any modification.
Mycoplasma contamination	We tested cell lines for mycoplasma contamination (MycAlert Mycoplasma Detection Kit, Lonza, LT07-418). All cell lines used in this work were free of mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	Experiments were performed using female mice that were 6-12 weeks old. 129S1 mice harboring Cre-activated conditional KrasLSLG12D/+Trp53flox/flox (KP mice) were used as an autochthonous mouse model of non-small cell lung cancer. Mice were fed with autoclaved food and water and maintained in ventilated cages in a light-dark cycle, temperature (18-23 °C), and humidity (40-60%) controlled pathogen-free vivarium at MGH.
Wild animals	The study did not involve wild animals.
Reporting on sex	We used thirteen (n = 13) female mice. Ten mice were tumor-bearing and three mice were tumor-free.

Field-collected samples	The study did not involve field-collected samples.
Ethics oversight	We conducted the animal study following the guidelines provided by the Institutional Animal Care and Use Committee (IACUC) of Massachusetts General Hospital (Protocol 2019N000133).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	<i>Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.</i>
Novel plant genotypes	<i>Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.</i>
Authentication	<i>Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.</i>